

Protected vacua, shared mediators and the alignment boundary

Singlet vacuum dynamics, canonical quark matching and explicit limits on flavor alignment

PerfectPower research monograph
5 October 2026

The selected model admits controlled vacua near ± 66 degrees. Its initialized equations receive perturbative supersymmetric protection; its energy selection, physical coefficient readout and quark couplings remain separately specified interactions.

Exact rational results and numerical scenario results are distinguished throughout. The worked construction is conditional and does not claim an independently derived theory of flavor.

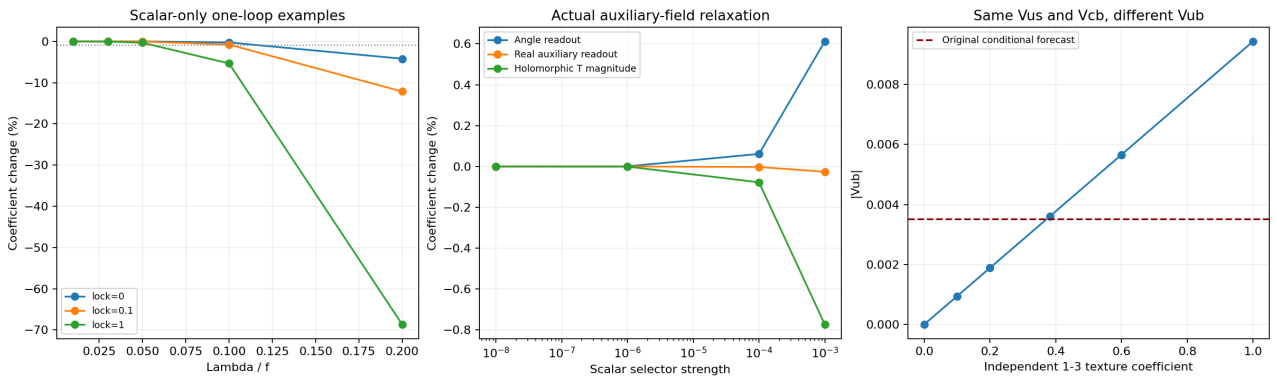
What this push accomplishes

The subsequent shared-mediator investigation is documented in B5_FLAVOR_MEDIATOR.md and included as the second part of the reader monograph. It derives exact canonical tree matching, establishes a scoped low-degree alignment obstruction, and tests an adjoint alternative. It transmits an initialized golden frame but does not derive that frame from symmetry.

The candidate now has an explicit renormalizable supersymmetric singlet sector, a holomorphic golden-coefficient extension, an actual auxiliary-field relaxation calculation, and a one-loop calculation including every scalar and fermion in that sector. The 34-field extension has 68 real scalar modes and 34 Weyl fermions. In the declared numerical examples, a small scalar energy selector prefers the branches near plus and minus 66 degrees, and the one-loop corrections leave that preference intact among all sixteen continued branches. The construction produces a quantitatively controlled realization of the nominated vacuum. It does not derive the initial superpotential coefficients or the energy-selector pattern from an independently specified symmetry, and it does not derive the nominated exact CKM relation from quark interactions.

The strongest negative result is exact: the scalar-only version of the potential is not closed under its own one-loop divergent counterterm. Polynomial division yields a nonzero torque remainder, and a Bezout identity shows that the remainder vanishes at none of the primitive real order-60 roots. Thus there is a specific radiative protection problem, even before quarks or gauge fields are introduced. The strongest constructive result is an initialized holomorphic constraint circuit whose supersymmetric zeros are perturbatively protected. Soft selection then produces small calculable departures instead of an exact unchanged angle.

The quark calculation exposes another independent issue. The declared flavor charges allow the required hierarchy of operators, but their coefficients are independent. Exact diagonalization of a selected triangular texture, even with its 1-3 coefficient initialized to the golden value, gives a different observable mixing strength. This is a constructive counterexample to assuming that the right spurion powers automatically imply the exact CKM formula.



The original exact carrier and energy winner

$$V'_{\text{lock}}(x) = P(x) [2\lambda P'(x) + \eta Q(x)]$$

Define $x=2 \cos(\theta)$, $P(x)=x^8-7x^6+14x^4-8x^2+1$ and $Q(x)=-1+3x-x^3$. The previously constructed potential obeys $V'(x)=P(x)Q(x)$. Its six nonconstant angular terms are $\cos(2\theta) + (2/3)\cos(3\theta) + (2/5)\cos(5\theta) + (1/4)\cos(8\theta) - (2/9)\cos(9\theta) - (1/6)\cos(12\theta)$. V differs from that angular expression by the constant $11/12$. Exact rational Sturm isolation and energy enclosures selected $x=2 \cos(66^\circ)$ on $[-2,2]$, with more than 0.060

separation from the next competing stationary energy in the polynomial normalization.

The new family $V_{\text{lock}}(x) = \lambda P(x)^2 + \eta V(x)$, $\lambda \geq 0$ and $\eta > 0$, obeys $V_{\text{lock}}'(x) = P(x)[2\lambda P'(x) + \eta Q(x)]$. Its global winner is unchanged: the added square is nonnegative, vanishes at the old winner, and the original potential already has a unique global winner in x . This statement covers the full declared parameter orthant without discretizing λ or η . The two full-circle CP partners remain degenerate. This is a mathematical family, not an assertion that the interaction generating λ or η has been identified.

The rational stationary-family constraint is stronger than stationarity under arbitrary real couplings at just one chosen angle. Rational coefficients force Galois-conjugate roots to share the polynomial factor. An isolated real linear torque condition at a prescribed angle is not equivalent to the eight rational relations found for the twelve-harmonic carrier. The field model must explain which of those structures it actually enforces.

A precise scalar-loop obstruction

$$h(x) = (4 - x^2)V''(x) - xV'(x), \quad \gcd(P, (h^2)') = 1$$

Give the angle the kinetic term $f^2 (\partial \theta)^2/2$ and the physical potential $\Lambda^4 V(2 \cos(\theta))$. With $D(x) = 4 - x^2$, its dimensionless angular curvature is $h(x) = D(x)V''(x) - xV'(x)$. The physical field-dependent squared mass is $\Lambda^4 h(x)/f^2$. The one-loop scalar divergence contains a coefficient proportional to this mass to the fourth power. Consequently the question of preserving the stationary family reduces to whether P divides the x derivative of h^2 .

For the original potential, the exact remainder is

$R(x) = 5040 - 6880x - 31360x^2 + 43600x^3 + 24160x^4 - 30080x^5 - 4400x^6 + 4800x^7$. Its gcd with P is one. The stored rational Bezout multipliers satisfy $a(x)P(x) + b(x)R(x) = 1$. Since $\sin(\theta)$ is nonzero at every primitive order-60 angle, a nonzero x torque also means a nonzero angular torque. This establishes that the original family is not closed under this particular divergent scalar counterterm. It does not establish that every model containing additional fields must fail.

The coefficient field can also be included. Write $c(x) = 1 - S_{12}(x)$, where $S_n(2 \cos(\theta)) = 2 \cos(n\theta)$, and take $F(\theta, C) = V_{\text{lock}}(2 \cos(\theta)) + \kappa[C - c(x)]^2$. On its tree valley, with kinetic scales f and g , the polynomial proportional to $\text{Tr}(M^4)/\Lambda^8$ is $[h + 2\kappa D(c')^2]^2/f^4 + 8\kappa^2 D(c')^2/(f^2 g^2) + 4\kappa^2/g^4$. The derivative is again reduced modulo P . All sixteen declared angular-only and two-scalar parameter cases give nonzero remainders. The calculation includes the coefficient scalar instead of silently integrating it out before evaluating the loops.

Finite scalar-loop displacements

$$\Delta V = \frac{m^4}{64\pi^2} \left[\log \frac{m^2}{\mu^2} - \frac{3}{2} \right], \quad \Delta \theta = -\frac{\Delta U'}{U''}$$

The scalar-only numerical calculation uses the local positive-mass $\overline{\text{MS}}$ expression $\Delta V = m^4 [\log(m^2/\mu^2) - 3/2]/(64\pi^2)$. The renormalized additional finite Wilson coefficients are initialized to zero at the stated μ . Every example declares Λ/f , the lock coefficient and μ/m_{tree} . Resetting that boundary condition at another μ defines another example; it is not a claim that a fixed physical theory depends on the arbitrary renormalization scale after consistent running.

At $\theta_0=66$ degrees, $U''=12.2092424208$ and $U'''=-140.847974475$. The first-order angular displacement is $-\Delta U'(\theta_0)/U''(\theta_0)$. For $\Lambda/f=0.1$, no added lock, and $\mu=m_{\text{tree}}$, the numerical stationary point is approximately 65.997446321 degrees. The angle-based coefficient changes by about -0.2663 percent. The small-ratio displacement scales as $(\Lambda/f)^4$. The derivative and curvature are independently checked against finite differences, and the nonlinear stationary solution is compared with its linear prediction.

The sweep retains sixty angular examples. One leaves the physical positive-depth CKM chart; it remains in the results with a null CKM forecast rather than being dropped. Eighty of eighty-one declared two-scalar examples produce a controlled local stationary solution with a positive numerical Hessian. The unsuccessful case is retained with its parameters and failure reason. These calculations certify neither a global quantum minimum nor a cosmological occupation probability.

At the tree portal $C=1-2 \cos(12 \theta)$, $dC/d\theta=22.8253563911$ at the selected angle. A one-percent coefficient tolerance corresponds locally to about 0.00958804 degrees. Adding a large classical locking term is not automatically a radiative solution: it changes the mass and its derivatives, which also changes its own loop torque. The stored sweeps illustrate this effect directly.

A cubic supersymmetric constraint circuit

$$z_n = z^n, \quad \Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$$

Introduce sixteen gauge-singlet chiral fields $Z_1, \dots, Z_{16}$, with $Z_1=Z$, and sixteen drivers $D_2, \dots, D_{16}, D_{\text{star}}$. All Z fields have R charge zero; all drivers have R charge two. The declared CP action is complex conjugation and the chosen coefficients are real. A common real coupling can multiply the superpotential. With a real positive reference mass M , the base superpotential is $W = \sum_{n=2}^{16} D_n (M Z_n - Z_{n-1}) + M D_{\text{star}} [Z_{16} + Z_{14} - Z_{10} - Z_8 - Z_6 + Z_2 + M]$. Every term has mass dimension three.

At $D=0$, F -flatness gives $z_n = z^n$, where $z_n = Z_n/M$, and $\Phi_{60}(z) = z^{16} + z^{14} - z^{10} - z^8 - z^6 + z^2 + 1 = 0$. There are sixteen simple complex roots. They have unit modulus and are the primitive order-60 roots. This is an explicit renormalizable circuit implementing the algebraic equation; it avoids introducing a degree-sixteen superpotential interaction in the fundamental variables.

The constraint Jacobian J has nonzero determinant at every root. Its determinant divided by $\Phi_{60}'(z)$ is minus one in the declared ordering. Thus the chosen vacua are isolated. The full holomorphic mass matrix has blocks $[[0, J^T], [J, 0]]$, and every singular value of J occurs twice. With canonical kinetic terms, each pair has four real scalar modes and four fermionic spin modes with equal masses. The unbroken-SUSY one-loop contribution cancels at the F -flat zeros.

Perturbative protection here is the Wilsonian superpotential non-renormalization result for an initialized, globally supersymmetric sector, with a nonsingular positive Kahler metric. It does not force the initial coefficient choices. Kinetic corrections do not remove a zero of all F terms, but they change spectra and matter when supersymmetry is broken. This distinction follows the Wilsonian treatment of non-renormalization, rather than treating a generic 1PI potential as an unchanged superpotential [2].

What the declared symmetry actually permits

For the base singlet sector, the complete holomorphic superpotential basis through field degree three has 2,448 monomials. R charge forces one driver, followed by zero, one or two neutral fields: $16(1+16+136)$. The selected circuit uses 37 monomials. CP and the declared R symmetry allow many other coefficients. They protect neither the chosen sparse pattern nor its relative integer values by themselves. The general CP-even quadratic Kahler metric contains 272 real parameters in the two neutral/driver blocks; the numerical mass calculations initialize it canonically.

A real change of the D_{star} tadpole changes $\Phi_{60}(z)$ to $\Phi_{60}(z)+\epsilon$. This deformation is allowed by the declared symmetries and changes both phase and radius. It demonstrates freedom in the initialized theory. It is not a claim that perturbative supersymmetric loops generate that absent term in the Wilsonian superpotential.

For a separate one-complex-field comparison, all CP-even operators $\text{Re}(z^a \bar{z}^b)$, $a \geq b$, $1 \leq a+b \leq 12$, are enumerated. There are 48 operators when no rotation symmetry is imposed; 42 have a nonzero carrier torque. The six desired harmonic numbers have gcd one. Among all cyclic rotations tested, orders two through sixty cannot permit every desired harmonic. This is a precise obstruction for that one-field CP-plus-cyclic class, not for arbitrary multi-field finite groups.

A holomorphic golden-coefficient extension

$$T = 2 - z^4 - z^6 + z^{14}, \quad T^2 - 3T + 1 = 0$$

The inverse phase needed for the original coefficient can be represented in the carrier without complex conjugation. Since $z^{60}=1$ on $\Phi_{60}=0$, $z^{-12}=z^{48}$. Exact reduction gives $T=1-z^{12}-z^{48}=2-z^4-z^6+z^{14}$ modulo Φ_{60} . It obeys $T^2-3T+1=0$. At the root with phase 66 degrees, $T=(3-\sqrt{5})/2$, with zero imaginary part.

Two additional chiral fields T and D_{gold} implement this linearly: $W_{\text{gold}}=M D_{\text{gold}}[T-2M+Z_4+Z_6-Z_{14}]$. T has R charge zero and D_{gold} has charge two. The extended model has 34 chiral fields and remains renormalizable. Its general allowed singlet superpotential through degree three contains 2,907 monomials. The extension fixes a golden spurion value in the selected initialized vacuum, while preserving the sixteenfold supersymmetric degeneracy. It does not fix a quark Wilson coefficient or explain why the twelfth power is the desired coefficient operator.

Actual field relaxation under energy selection

To lift the degeneracy, the worked scalar deformation is $V_{\text{soft}}/M^4=g_W^2 \epsilon \sum a_n \text{Re}(z_n)$, with the six nonconstant Fourier coefficients a_n . This is an explicitly chosen supersymmetry-breaking scalar tadpole pattern. At exact circuit roots its leading energy is the original angular potential times $g_W^2 \epsilon$. The full fields are then allowed to relax; they are not constrained to remain equal to powers of a unit-modulus number.

For fixed z_1 , the other fifteen base neutral fields enter the F-term potential quadratically. They are eliminated by linear least squares, including the scalar tadpoles. The remaining complex z_1 is optimized in both its real and imaginary coordinates. All sixteen branches are continued for $\epsilon=10^{-8}, 10^{-6}, 10^{-4}, 10^{-3}$. The branches descended from root powers eleven and forty-nine have the lowest energies in every declared continuation. Every retained branch has a positive reduced tree Hessian. This is a comparison of local continued vacua, not a proof that the

full polynomial field potential has no distant competing vacuum or runaway.

At $\epsilon=10^{-4}$, the positive branch has phase 66.0005874177 degrees and radius 1.00002214970. The angle-only readout gives $C=0.382200029974$. The nonholomorphic real auxiliary readout $1-2 \operatorname{Re}(z_{12})$ gives 0.381955432748. The holomorphic extension gives $T=0.381669772701-0.000178406720 i$. These are different operators once the fields relax. A quark interaction must specify which operator it actually uses. Choosing a real auxiliary readout in a holomorphic superpotential would require a different construction; it cannot be substituted silently.

Full-circuit leading quantum calculation

$$A = J^\dagger J, \quad B = \sum_k \overline{F_k} H_k$$

The full quantum calculation includes the 34 chiral fields of the golden extension. On the R-symmetric $D=0$ slice, let F be the vector of driver constraints, J their Jacobian with respect to the seventeen neutral fields, and H_k their constant quadratic Hessians. Define $A=J^\dagger J$ and $B=\sum \operatorname{conjugate}(F_k) H_k$. In canonical $\sqrt{2}$ real and imaginary neutral coordinates, the scalar squared-mass matrix is $[[\operatorname{Re}(A+B), -\operatorname{Im}(A)-\operatorname{Im}(B)], [\operatorname{Im}(A)-\operatorname{Im}(B), \operatorname{Re}(A-B)]]$. The driver scalar block has the corresponding unsplit singular-value spectrum.

The driver scalar trace cancels half of the fermion trace. The complete singlet one-loop potential therefore equals $g_W^4/(64 \pi^2)$ times $[\sum_{(34 \text{ neutral real modes})} b_i^2 (\log(g_W^2 b_i/\mu^2)-3/2)-2 \sum_{(17 \text{ Jacobian modes})} a_i^2 (\log(g_W^2 a_i/\mu^2)-3/2)]$. This expression accounts for all 68 real scalars and 34 Weyl fields, not just the light angular mode. It is evaluated in a DRbar example with canonical renormalized Kahler metric and specified scalar tadpoles at the reference scale. There are no gauge fields in this singlet-sector calculation.

The gradient uses eigenvector contractions with the analytic derivatives of A and B . It is independently checked against potential differences. At exact F-flat roots, the potential and gradient cancel to numerical roundoff. At the softly selected vacuum, the gradients need not cancel. The quantum stationary equations are solved in all thirty-four real neutral coordinates; the corrected neutral-sector Hessian is numerically positive. An independent expression for the complete 68-scalar, 34-Weyl effective potential also computes the quantum driver-block Hessian by finite differences. The unbroken common driver R phase makes the neutral-driver block vanish at zero drivers. Both corrected blocks have positive minimum eigenvalues in all nine declared cases. Step-size comparison checks the driver Hessian, while the agreement of the independent full and reduced potentials checks the mode counting. These are numerical local stability checks at one loop, rather than exact certificates or a higher-loop theorem.

Nine epsilon/coupling examples are retained: $\epsilon=10^{-6}, 10^{-4}, 10^{-3}$ and $g_W=0.1, 0.3, 1$. All produce controlled numerical neutral-sector minima. Leading tree-plus-one-loop energy comparisons among all sixteen continued branches retain root powers eleven and forty-nine. At $\epsilon=10^{-4}$ and $g_W=0.3$, the phase becomes 66.0005860473 degrees, a quantum displacement of approximately -1.37037 times 10^{-6} degrees relative to the relaxed tree minimum. The coefficient field becomes $T=0.381670086337-0.000178010587 i$. The largest actual neutral-coordinate quantum displacement is about 4.67 times 10^{-7} . These are chosen weak-coupling examples, not experimental uncertainties or a derived cosmological scale.

Quark operators: hierarchy without an enforced coefficient

The declared holomorphic quark sector uses two flavor charges. The left-handed doublets Q_i have charges (1,1),(0,1),(0,0), while u_i^c and d_i^c have the opposite charges. H_u, H_d and all neutral circuit fields have zero flavor charge. A has charge (-1,0), B has charge (0,-1), and Q, f^c have R charge one. The scalar A and B backgrounds remain free inputs. No anomaly-free gauged flavor completion or mechanism fixing their magnitudes is claimed.

All allowed operators $Q_i H f_j^c A^a B^b$ times neutral monomials are enumerated through superpotential field degree six. There are 6,460 operators with the sixteen base neutral fields, and 7,560 when T is included. The up and down sectors have independent real coefficients before spontaneous CP breaking. The pure singlet enumeration and this Yukawa enumeration are complete within their declared sectors and degree bounds; they are not an enumeration of all Standard Model, lepton or higher-Kahler interactions.

In particular, $Q_1 H d_2^c A$, $Q_2 H d_3^c B$, and $Q_1 H d_3^c A B$ carry independent coefficients. Allowing their spurion powers does not relate those coefficients. A T insertion can multiply the third operator, but its independent coefficient and the lower-degree $A B$ operator remain allowed. Neutral fields can also appear in diagonal and up-sector entries. The phase of the scalar vacuum therefore does not automatically become the standard CKM phase. Higher-Kahler interactions such as $Q_1^\dagger Q_2 A^\dagger/M$ are allowed and can alter canonical normalization.

A derived texture counterexample

$$V_{\text{CKM}} = U_u^\dagger U_d, \quad U_f^\dagger Y_f Y_f^\dagger U_f = \text{diag}(y_f^2)$$

The worked trial uses $Y_u = \text{diag}(u_1, u_2, u_3)$ and an upper-triangular Y_d with entries $Y_{12} = a d_2$, $Y_{23} = b d_3$ and $Y_{13} = c a b \exp(-i \theta) d_3$. The diagonal spectra are declared positive numerical inputs. This is a selected member of the allowed operator family. Up-sector off-diagonal terms and most neutral insertions are initialized to zero; the symmetry does not forbid them.

The matrices are diagonalized through the eigenvectors of $Y Y^\dagger$. For each independent c , only a and b are matched to the two existing anchors $|V_{us}| = 0.22431$ and $|V_{cb}| = 0.0411$. The values of $|V_{ub}|$ and CP observables are then consequences of the matrix, without fitting them. At $c = \phi^{-2}$ and $\theta = 66$ degrees, the texture gives $|V_{ub}| = 0.00360299463$ and effective depth 0.390811666690, rather than the nominated chart's 0.00352144218 and 0.381966011250. Its standard CKM phase is approximately 65.9786293 degrees. Finite diagonalization effects matter even in this simple hierarchical matrix.

Changing the independent coefficient while retaining the same two anchors changes $|V_{ub}|$ substantially. The stored six-coefficient sweep supplies explicit examples. This is a mathematical obstruction to inferring the exact relation $s_{13} = C s_{12} s_{23}$ from charge arithmetic alone. A stronger quark construction must relate Wilson coefficients, kinetic normalization and matrix diagonalization, or predict the correction instead of absorbing it into a fit.

Scale evolution as a separate matching test

$$16\pi^2 \frac{dY_u}{d \log \mu} = \left[\frac{3}{2} (H_u - H_d) + \alpha_u I \right] Y_u$$

The implementation also integrates one-loop Standard Model matrix Yukawa equations using $t=\log(\mu)$, left-row Yukawa convention and SU(5)-normalized g_1 . In this convention $\beta_u=[(3/2)(H_u-H_d)+T_{\text{gauge},u}]Y_u/(16\pi^2)$, with the corresponding down and lepton equations and the one-loop gauge equations. The source's $\log(\mu^2)$ convention is converted explicitly [3]. Rephasing invariance, a common-left-basis covariance check and the top-only limit independently check the implementation.

The example initializes the original exact CKM chart with declared illustrative spectra and gauge couplings. At $\mu/\mu_0=10^8$, its effective depth differs from ϕ^{-2} by about -8.62 times 10^{-5} relatively, and its standard phase differs by about -0.00349036 degrees. The absolute $|V_{ub}|$ also changes as the anchors run. No measured matching scale is supplied. This is a standalone SM evolution diagnostic, not a threshold-matched evolution of the supersymmetric circuit. It shows that a claim of an exact flavor relation must identify the scale at which it is imposed.

What is now supported and what remains physical

The exact supports are the carrier algebra, the stationary-family identities, the global-winner preservation argument, the scalar-counterterm remainder and Bezout witness, and the complete declared finite operator spaces. The supersymmetric protection is a conditional application of the established Wilsonian non-renormalization result to the supplied globally supersymmetric circuit. The mass spectra, relaxed fields, one-loop extrema, quark diagonalizations and SM evolution are numerical computations with explicit parameters and limitations. No new Lean theorem is claimed.

The constructed model answers whether an initialized protected equation can be implemented by renormalizable fields and whether a specified weak selector produces a controlled nearby vacuum. It does so. The outstanding physics is the origin of the sparse initialized coefficients and selector terms, a quark interaction enforcing or replacing the nominated exact mixing rule, consistent kinetic and threshold matching, higher-loop stability, and a vacuum history. The branch comparison does not choose the positive CP sign. A complete theory of strong CP or baryogenesis has not been supplied.

Reproduction and retained calculations

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_completion.py` from the repository root, then the corresponding `plot_flavor_completion.py`. Run unittest discovery with `test_flavor_completion.py` for the new mathematical and numerical checks, or `test_flavor*.py` for the combined flavor checks. NumPy and SciPy support numerical spectra, extrema and running; Matplotlib supports the scientific figure. Exact polynomial identities and operator labels use rational arithmetic and the existing PerfectPower algebra.

The receipts retain all declared cases, including one uncontrolled two-scalar continuation and one angular example outside the positive-depth CKM chart. The two exhaustive quark-operator tables are stored as deterministic gzip-compressed JSON; ordinary gzip decompression recovers the complete tables. The manifest covers the scientific JSON outputs, including those compressed tables. Fresh-copy reproduction checks those outputs in the same numerical runtime. A byte-identical replay is evidence of reproducibility in that environment; the tests provide separate mathematical and numerical checks. PDF generation uses ReportLab and standard installed fonts. Source code, prior repository work, raw receipts, diagrams and the monograph are included in the complete handoff archive.

The preceding vacuum-completion run completed 841 tests successfully, with four skips. Its 25 new completion tests cover the exact carrier obstruction, circuit elimination and mode pairing, independent loop gradients, corrected driver Hessian, complete declared operator enumeration, texture counterexamples and scale-evolution covariance. A fresh archive extraction reproduced all twenty scientific receipt, manifest and figure files byte for byte in the declared runtime. No new Lean theorem was added in that push.

Primary references

- [1] S. Coleman and E. Weinberg, Radiative Corrections as the Origin of Spontaneous Symmetry Breaking, *Physical Review D* 7, 1888 (1973). <https://doi.org/10.1103/PhysRevD.7.1888>. This supplies the effective-potential framework; the worked scalar and circuit schemes are stated explicitly above.
- [2] N. Seiberg, Naturalness Versus Supersymmetric Non-renormalization Theorems (1993), <https://arxiv.org/abs/hep-ph/9309335>. The relevant distinction is protection of a chosen Wilsonian superpotential, with separate kinetic and supersymmetry-breaking effects. It does not derive the chosen flavor coefficients.
- [3] A. V. Bednyakov, A. F. Pikelner and V. N. Velizhanin, Three-loop SM beta-functions for matrix Yukawa couplings (2014), <https://arxiv.org/abs/1406.7171>. The one-loop matrix conventions and scale conversion are the parts used here.
- [4] I. de Medeiros Varzielas, D. Emmanuel-Costa and P. Leser, Geometrical CP violation from non-renormalisable scalar potentials (2012), <https://arxiv.org/abs/1204.3633>. This is a comparison for symmetry-constrained higher-order operators preserving particular vacua; it does not establish the present order-60 construction.

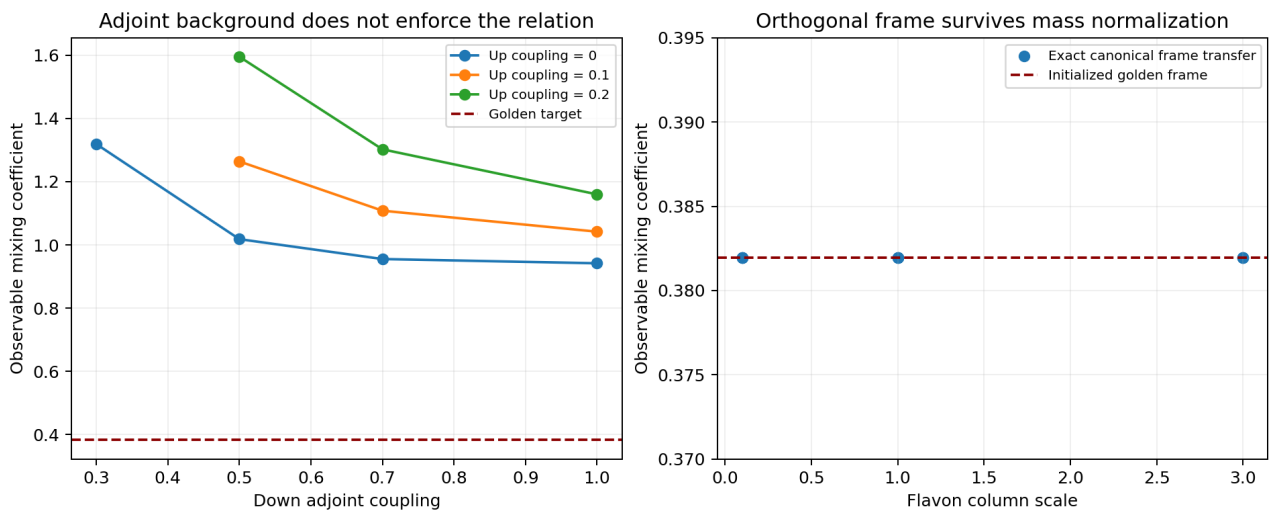
Shared mediators and the alignment boundary

What this mediator push establishes

The next step now has an explicit non-Abelian quark mediator construction and an exact tree matching calculation that includes the light right-handed kinetic normalization. The family symmetry makes each heavy mediator mass proportional to the identity and gives one contraction for each flavon column. Separately orthogonal flavon frames then determine the CKM matrix independently of the six column magnitudes and the two flavor-universal mediator masses. This is a constructive quark interaction that transmits an initialized mixing frame without the finite-mass diagonalization errors of the previous triangular texture.

It does not independently derive that frame. The accompanying scalar analysis gives a precise obstruction to the simplest alignment route. For six fundamental flavons with separate phase symmetries, fixed positive norms and exactly orthogonal up and down frames, every CP-even orientation term through scalar degree six reduces to a constant plus a linear function of $|V_{ij}|^2$. An exact rational assignment certificate proves that a permutation always attains its global minimum on the frame manifold. A generic nondegenerate cost therefore cannot select the required hierarchical CP-violating mixing. Degenerate costs can leave mixed flat directions; those are not a unique prediction.

I also tested a more favorable alternative in which an adjoint mediator mass contains initialized neighboring mixings, a golden 1-3 entry and the 66-degree phase. Both up and down adjoint couplings are allowed independently by the symmetry. Exact tree matching and canonical normalization were performed before extracting CKM quantities. Thirty of thirty-six declared cases satisfy the numerical control criteria. Every accepted case fails the golden target by more than one percent, despite matching the two existing anchors. The six unsuccessful or uncontrolled cases are retained with their reasons. The construction is therefore a tested candidate and a boundary result, not a completed predictive theory of flavor.



The exact observable target

$$C_{\text{eff}} = \frac{w(1-w^2)}{uv} = \phi^{-2}$$

Write $u=|V_{us}|$, $v=|V_{cb}|$ and $w=|V_{ub}|$. The nominated physical relation is $C_{\text{eff}}=w(1-w^2)/(uv)=\phi^{-2}$, where $\phi=(1+\sqrt{5})/2$. This follows from the original standard-angle rule $s_{13}=C s_{12} s_{23}$ because $u=s_{12} c_{13}$, $v=s_{23} c_{13}$ and $w=s_{13}$. It is already rephasing invariant. Relating a Yukawa matrix entry to other entries is insufficient unless its diagonalization yields this observable equation.

For $p=u^2$, $q=v^2$ and $r=w^2$, define $A=r(1-r)^2$. Eliminating C from $C^2-3C+1=0$ gives $A^2-7pqA+(pq)^2=0$. The positive physical branch $C_{\text{eff}}<1$ selects ϕ^{-2} ; the other positive root is ϕ^2 . The equation does not fix a CP phase. In raw fixed-norm flavon overlaps its formal maximum scalar degree is twenty-four. Squaring it into an energy term would encode the desired result by construction and would not explain its origin.

The existing anchors remain $u=0.22431$ and $v=0.0411$. The small-branch target is $w=0.00352144218276$. No new experimental data or fit to $|V_{ub}|$ or CP measurements was introduced. All numerical examples use declared illustrative quark spectra and dimensionless mediator units. A claim of an exact relation still needs a specified matching scale and subsequent threshold and running calculations.

An exact Abelian obstruction for the initialized circuit

Consider an additional ordinary diagonal Abelian phase action preserving every monomial of the initialized singlet superpotential, with its mass M and numerical coefficients neutral. The recursion terms force $q(Z_n)=n q(Z_1)$ and $q(D_n)=-n q(Z_1)$. The constant D_{star} term forces $q(D_{\text{star}})=0$. Its remaining terms require $n q(Z_1)=0$ for $n=2,6,8,10,14,16$. Their greatest common divisor is two, so the general character relation is $2 q(Z_1)=0$.

The golden extension has a constant D_{gold} term, forcing $q(D_{\text{gold}})=0$. Its linear $D_{\text{gold}} T$ term then forces $q(T)=0$. Hence an operator O and $O T$ have the same additional charge. Such a charge rule cannot forbid $Q_1 H d_3^c A B$ while allowing $Q_1 H d_3^c A B T$. The implementation derives the monomial constraints from the existing circuit rather than a separate assumed polynomial, and checks cyclic groups of orders one through 120. The focused test independently extends those checks through order 240.

This result is deliberately restricted. Charged mass or coupling spurions, symmetries mixing fields, and non-Abelian representations can change the premises. Treating the initialized coefficients as transforming spurions would supply a different model and would require its own alignment and coupling audit. The obstruction does not rule out flavor symmetry in general.

The declared non-Abelian mediator sector

Use the Standard Model gauge group, a formal global $SU(3)_F$ family symmetry, six independent $U(1)$ flavon phase symmetries and CP. Q is a family anti-triplet. The three right-handed-conjugate up fields and the three right-handed-conjugate down fields are family singlets. There are six complex gauge-singlet fundamental flavons F_{ui} and F_{di} . Each carries its own phase charge $+1$, while its associated right-handed-conjugate quark carries -1 . The Higgs and Q have zero phase charge.

Introduce heavy X_u and X_d family anti-triplets and X_u^c and X_d^c family triplets. X_u and X_u^c are vectorlike color-triplet weak-singlet fermions with hypercharges $+2/3$ and $-2/3$. X_d

and X_d^c have hypercharges $-1/3$ and $+1/3$. Each family triplet gives three heavy Dirac species, for six species overall. These pairs are vectorlike under the Standard Model. The global family group is not claimed to be an anomaly-free gauged completion; its Goldstone and cosmological questions are outside this calculation.

In ordinary left-handed Weyl notation, the allowed core interactions are $M_f X_f \cdot X_f^c$, $h_f Q \cdot H_f X_f^c$ and $\sum_i \lambda_{fi} X_f \cdot F_{fi} f_i^c$, plus their Hermitian conjugates. H_u is the Higgs doublet and H_d is its conjugate doublet. This notation is an ordinary non-SUSY mediator model. It is not a combined softly broken completion of the previously constructed 34-chiral-field singlet sector.

The operator audit exhausts fermion mass bilinears and two-fermion-one-scalar Yukawa invariants in this declared quark sector. It checks Standard Model gauge representations, family representations and all six phase charges. With the seventeen neutral scalar backgrounds Z_1 through Z_{16} and T , including their conjugates, there are 78 independent invariant coefficients. Two are bare universal masses, two are Higgs vertices, six are flavon-column vertices and 68 are neutral-scalar universal-mass vertices. CP permits real coefficients; it does not equate them. Lepton interactions and a complete scalar potential for all neutral backgrounds are outside this finite audit.

Adding one complex phase-neutral family adjoint Σ and its conjugate adds four invariants, giving 82. They are $X_u \Sigma X_u^c$, $X_u \Sigma^\dagger X_u^c$, $X_d \Sigma X_d^c$ and $X_d \Sigma^\dagger X_d^c$. Up and down coefficients are independent. In the Hermitian background used below the two coefficients in each sector combine into one real number, but the symmetry does not impose any relation between the sectors or between an adjoint coupling and a singlet mass.

Canonical tree matching without a small-flavon expansion

$$A = M^{-1}L, \quad K = I + A^\dagger A, \quad Y = -hAK^{-1/2}$$

Let L_f be the 3 by 3 matrix of flavon columns multiplied by their λ_{fi} couplings, and let M_f be the invertible heavy mediator mass matrix. At zero Higgs background the heavy fermion row block is $[L_f, M_f]$. Define $A_f = M_f^{-1} L_f$, $K_f = I + A_f^\dagger A_f$ and $R_f = K_f^{-1/2}$. A normalized basis for the three light right-handed fields is $N_f = [R_f; -A_f R_f]$. Direct multiplication gives $[L_f, M_f]N_f = 0$ and $N_f^\dagger N_f = I$.

The Higgs vertex projected into that normalized null space gives $Y_f = -h_f A_f R_f$. This is exact at tree order at unbroken electroweak matching, to all orders in the ratio of flavon mixing to heavy mass. It is not the unnormalized expression $-h_f M_f^{-1} L_f$. No loop threshold is included. At finite Higgs vacuum value, higher-dimensional Higgs operators and mixing with weak-singlet heavy fermions also affect the charged current. The dimension-four effective CKM matrix is unitary; the finite-Higgs full theory need not give an exactly unitary light 3 by 3 submatrix.

The focused checks verify the normalized null frame and Yukawa projection directly. They also compare against independent singular values of the complete 6 by 6 fermion mass matrix $[[0, h_f v_H I], [L_f, M_f]]$. The three light singular values divided by v_H converge to those of the matched Y_f . A separate finite-Higgs test compares the light charged-current overlaps with the matched CKM limit and checks their small departure from unitarity. Noncommuting mass matrices are tested under a common family-basis rotation, so the checks do not rely only on diagonal mediator masses.

What universal masses can and cannot transmit

For a universal mass $M_f = m_f I$ and a general singular decomposition $L_f = U_f \text{diag}(s_f) W_f^\dagger$, the canonical Yukawa matrix has the same left eigenvectors U_f . Its singular values are $|h_f| s_f / \sqrt{|m_f|^2 + s_f^2}$. This map is strictly increasing for positive s_f , preserving nondegenerate mass ordering. Thus changing a flavor-universal mediator mass, even by a complex golden or cyclotomic background, cannot change the mixing angles or physical CKM phase of fixed L_u and L_d . This statement does not require orthogonal flavon columns.

With separately orthogonal columns, $L_f = U_f \text{diag}(\ell_f)$, the stronger separation holds: changing each individual column magnitude only changes its associated quark singular value. While retaining the assigned nondegenerate mass ordering, the CKM matrix is $U_u^\dagger U_d$ and is independent of those magnitudes and the universal masses. Twelve declared examples initialize $U_u = I$ and U_d from the original golden CKM chart, then vary column magnitudes and real or complex universal masses. They retain the nominated coefficient and 66-degree standard phase to numerical precision. These are conditional transfer checks. The down frame was copied from the target chart and has not been derived from a symmetry-controlled potential.

A separate three-case test uses nonorthogonal random columns and changes only universal masses. It confirms the general unchanged-eigenvector result while the spectra change. Both results narrow the physical role a golden field can play: placing it only in a universal heavy mass does not orient the CKM matrix. Placing its phase as an overall Yukawa-column multiplier also fails, because $(Y \text{diag}(\exp(i \theta_j)))(Y \text{diag}(\exp(i \theta_j)))^\dagger = Y Y^\dagger$. A physical CP phase must influence relative left-handed directions or interfering flavor structures.

The complete low-degree triplet alignment basis

For the six complex triplets alone, write $G_{ab} = F_a^\dagger F_b$. Independent phase invariance requires each triplet label to have equal occurrences with and without conjugation. At scalar degree two there are six norms. At degree four there are 21 products of norms and fifteen squared overlaps $|G_{ab}|^2$, giving 36 quartic invariants. Including the quadratic terms, the renormalizable triplet potential has 42 independent real coefficients. One Standard Model Higgs adds its quadratic term, its quartic term and six Higgs-norm/flavon-norm portals, giving fifty terms in that restricted scalar sector.

At degree six the CP-even basis has 56 triple products of norms, ninety norm-times-squared-overlap terms and twenty real Gram triangles $\text{Re}(G_{ab} G_{bc} G_{ca})$ with distinct labels. Products of balanced $SU(3)$ epsilon tensors are Gram determinants and reduce to this basis. Unbalanced epsilon contractions are forbidden by the six separate phase symmetries. The first Gram-rank determinant relation for six triplets in three dimensions has degree eight, so it does not remove any of these degree-six contraction types.

Now restrict to positive fixed norms and exactly orthogonal U_1, U_2, U_3 and D_1, D_2, D_3 frames. Overlaps within each frame vanish. Every three-label triangle contains two labels from one frame, so every Gram triangle vanishes. Norm products are constant. The surviving squared cross overlaps are proportional to $B_{ij} = |U_i^\dagger D_j|^2 = |V_{ij}|^2$. Norm factors at degree six only modify their coefficients. Therefore the complete orientation dependence through degree six is $E = \text{constant} + \sum_{ij} K_{ij} B_{ij}$.

Fixed norms and exact separate orthogonality are assumptions of this theorem. Arbitrary finite quartic coefficients in the full potential can distort the frames, and a model outside this regime requires a new calculation. An adjoint introduces new orientation tensors and is also outside this theorem. Neutral singlets can change norm-dependent parameters; multiplying a quartic

overlap by a singlet would exceed renormalizable degree and does not create a new degree-four orientation tensor.

A rational global certificate for the orientation obstruction

$$E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij} \geq E_{\text{permutation}}$$

For any unitary relative frame, B is doubly stochastic: its entries are nonnegative and each row and column sums to one. For each declared rational cost matrix K , the implementation finds row and column numbers α_i and β_j with $\alpha_i + \beta_j \leq K_{ij}$. It also finds a permutation attaining the same sum $\sum_i \alpha_i + \sum_j \beta_j$. Define reduced costs $R_{ij} = K_{ij} - \alpha_i - \beta_j$, which are exactly nonnegative rational numbers.

Then $E = \sum_i \alpha_i + \sum_j \beta_j + \sum_{ij} R_{ij} B_{ij}$ is bounded below by the permutation energy. A permutation matrix is itself unitary, so the bound is attained within the physical frame manifold. The receipt supplies the rational dual numbers, reduced costs and all winning permutations. Tests check eighty further rational cost matrices independently against all six assignments and compare the identity against nontrivial random unitaries.

This proves that a permutation is always a global minimizer of the restricted linear orientation energy. With a unique minimum assignment, no nonpermutation B attains the same energy. For degenerate costs, a mixed CP-violating frame can instead belong to a flat minimum set. The constant-cost example explicitly retains such a direction. The correct conclusion is that this potential cannot uniquely select the nominated nonpermutation CKM matrix in the stated regime, not that every possible minimum must have zero CP violation.

The first higher-degree CP capability example

Scalar degree eight permits $|F_{ui}^\dagger F_{dj}|^4$ and real four-overlap cycles. With equal fixed norms, the chosen positive sum of fourth powers reduces to $\sum_{ij} B_{ij}^2$. Each row obeys $\sum_j B_{ij}^2 \geq 1/3$, so the total is at least one. Equality requires every $B_{ij} = 1/3$. The unitary three-dimensional Fourier frame reaches the bound and has $|J| = 1/(6 \sqrt{3})$, with a conjugate CP partner.

This is an explicit CP-even orientation interaction that can select nonzero physical $|J|$ when paired with nondegenerate quark column magnitudes. It selects democratic mixing, with every CKM magnitude $1/\sqrt{3}$, and fails the hierarchical target. Its coefficient pattern is chosen; other degree-eight invariants are independently allowed under the declared symmetry. No renormalizable UV generation of that pattern or connection to the protected 66-degree singlet vacuum has been provided. The example demonstrates what new orientation dependence becomes possible at degree eight, while leaving the actual golden alignment unresolved.

The favorable adjoint candidate and its failure

The numerical candidate uses a Hermitian traceless adjoint background Σ with entries $\Sigma_{12}=a$, $\Sigma_{23}=b$ and $\Sigma_{13}=\phi^{-2} a b \exp(-i \theta)$, together with their conjugates and zero diagonal. θ is initialized at 66 degrees. The mediator masses are $M_u = I + g_u \Sigma$ and $M_d = I + g_d \Sigma$. The columns before filtering are diagonal, using the same illustrative hierarchies as the preceding quark calculation. This gives the golden factor and phase a direct place in a flavor-dependent mediator, which a universal mass could not supply.

The assumed adjoint background is deliberately favorable and explicitly initialized. No scalar potential producing it is claimed. Both g_u and g_d are independent allowed couplings. Canonical tree matching is performed for each sector, followed by exact numerical diagonalization of $YY^\dagger$. Only a and b are matched to $|V_{us}|$ and $|V_{cb}|$. They are not tuned to $|V_{ub}|$ or any CP observable.

The declared sweep uses $g_u=0,0.1,0.2$; $g_d=0.3,0.5,0.7,1$; and column scales $0.1,1,3$. A case is accepted only when the two anchor residuals are below 10^{-9} , both mediator matrices have minimum singular value at least 0.2 and $\max(a,b) \leq 1$. Thirty cases are accepted and six fail the matching or control criteria. Accepted C_{eff} ranges from approximately 0.941676 to 1.595694 , against the golden target 0.381966 . All accepted cases fail the target at the one-percent level. The input phase also does not become the standard CKM phase automatically. Four conjugate-pair checks retain magnitudes and reverse J .

Unnormalized Yukawa predictions are retained for the same backgrounds so that the kinetic correction can be inspected. They are not substituted for the canonical result. The failure is not a theorem against all adjoint models; it demonstrates that this favorable entry pattern and the allowed shared-mediator couplings do not enforce the desired observable relation. A successful adjoint construction would need a derived vacuum and additional relations among its allowed couplings, followed by the same normalization and mixing checks.

Reproduction, validation and the next interaction target

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_mediator.py`, followed by `python python/plot_flavor_mediator.py` with the same environment. The runner uses only the existing $|V_{us}|$ and $|V_{cb}|$ means. It retains exact charge, operator and rational dual certificates, the conditional transfer cases, all adjoint attempts and the scientific figure. NumPy and SciPy support the numerical matching; the exact charge and assignment statements use integer or rational arithmetic. There are no new Lean theorems or newly acquired experimental datasets.

The twenty focused tests cover the charge obstruction, independent operator-list construction, exact rational assignment witnesses, canonical null-space matching, the complete fermion mass comparison, family-basis covariance, CP conjugation, universal-mass invariance and the failed golden enforcement. They accompany the repository-wide suite and a fresh-extraction byte comparison of the retained scientific artifacts. The integrated repository-wide run completed 879 tests successfully, with four skips. A fresh archive extraction reproduced all 32 scientific receipt, manifest and figure files from both flavor investigations byte for byte. The full test and replay results are recorded in the validation receipt for this push.

The resulting research target is more specific than adding a golden Yukawa coefficient. The quark mediator interaction can already transmit an orthogonal frame exactly, so the remaining positive construction is a symmetry-controlled relative frame that produces the golden amplitude relation and the physical CP phase. Within the tested phase-symmetric triplet class, a uniquely selected nonpermutation frame requires new orientation dependence beyond degree six or a change of fields or assumptions. Higher-degree interactions, additional non-Abelian representations or nonorthogonal-frame dynamics are possibilities to test; none is asserted here to solve the problem. Threshold matching, running, a consistent combined singlet/flavor theory and a vacuum history remain separate requirements.

Primary comparisons and their limits

S. Antusch, S. F. King, M. Malinsky and M. Spinrath, Quark mixing sum rules and the right unitarity triangle, <https://arxiv.org/abs/0910.5127>. Its texture-zero sum-rule route motivates deriving mixing from both Yukawa sectors. It does not establish the present golden relation.

R. Alonso, M. B. Gavela, L. Merlo and S. Rigolin, On The Potential of Minimal Flavour Violation, <https://arxiv.org/abs/1103.2915>. Its analysis provides context for restrictions of low-degree dynamical flavor potentials. The six-triplet fixed-frame obstruction above has separate explicit premises and its own rational certificate; no claim of duplicating every model in that paper is made.

J. Gehrlein, J. P. Oppermann, D. Schafer and M. Spinrath, An $SU(5) \times A_5$ Golden Ratio Flavour Model, <https://arxiv.org/abs/1410.2057>. It provides an example of an explicit renormalizable flavor and messenger construction. Its golden mixing result concerns the neutrino sector, not our nominated CKM coefficient. Only the primary abstracts and metadata were used for these comparisons.

Recovered M22 geometry and the flavor frame

What was recovered

The repository preserves selected original files from WILSON_FORMATION_FORGE v0.26.0, including its M22/A5 Wilson algebra, Schur-selector theorems, transport-category results, and a 1,023-word exact integer ledger. The provenance manifest is `galois_merge/receipts/vendor_sources.json`. This is a verified subset of the old work, not recovery of every later Atlas or EQ-LAB release. The saved ledger has SHA-256 `1351e514fcb5f23832f65665e0a12d2e172f8131c34875de0e03cddc5aec3093`. The large original orbital tensor is absent from the subset; the ledger is nevertheless sufficient to reconstruct its two-generator algebra faithfully.

The carrier is $V = C[M22/A5]$, of dimension 7,392. Its commutant $H = \text{End}_{M22}(V)$ has dimension 152. The complex simple block sizes are 1,3,3,2,5,5,5,3,3,6. Their squares sum to 152. Thus there are four three-dimensional multiplicity carriers available for a frame calculation. They are multiplicity spaces of larger M22 representations, rather than four nontrivial three-dimensional representations of M22.

The recovered Q192 generators are $X = A49$ and $Y = -A12 - A48 - A59 + A95 + A116 + A125$. X is self-adjoint and Y is skew-adjoint. This convention differs from the earlier Q7 signed operator; no Q192 calculation here is silently identified with Q7 or with its exceptional 154 block. Likewise, the three-dimensional minus-context factor in the separate 486-dimensional transport algebra is not identified with a Wilson multiplicity carrier or a physical family triplet.

Exact reconstruction from the word ledger

The new verifier `python/develop_m22_frame.py` selects 152 independent ledger rows through word degree eight, inverts their integer coordinate matrix over the rationals, and reconstructs left and right multiplication by X and Y . It also reconstructs the orbital transpose permutation from reversed words and their skew signs. No CKM number, target phase, or golden coefficient enters this reconstruction.

Every available one-step left and right extension of all 511 words through degree eight is checked exactly against the original ledger. The transpose squares to the identity and is an actual orbital permutation. All four left/right generator commutators vanish exactly. Adjointness recovers positive integral orbital subdegrees, normalized at the identity; their sum is exactly 7,392. The recovered regular matrices are integral, not numerical fits.

The exact characteristic polynomial of X is $(t-30)^9(t-3)^{51}(t+2)^{65}(t+5)^{27}$. The exact Y factorization is recorded in `receipts/m22_frame/frame_audit.json`. In particular it contains the factors t^2+1452 and t^2+2732 relevant to the two real three-dimensional carriers, and a degree-six factor for the conjugate complex pair. Regular-module multiplicities and physical permutation-module multiplicities are different; only the shared root sets are being used here.

Unitary one-copy carrier extraction is a separate numerical step. A generic Hermitian left-multiplication selector commutes with right multiplication, so its eigenspaces supply right-invariant carriers. Word traces through degree five distinguish the conjugate complex blocks. Even trace data alone would merge that pair incorrectly. The extracted block sizes

reproduce all ten expected simple blocks. The largest reported invariance residual is below $3.3\text{e-}11$. Carrier coordinates and numerical overlap matrices are explicitly not promoted to exact algebraic coordinates.

Why equivariance alone cannot select the frame

Write V as the direct sum of R_λ tensor M_λ , where R_λ is an irreducible M_{22} carrier and M_λ is its multiplicity space. M_{22} acts as $\rho_\lambda(g)$ tensor I on this summand. Every operator I tensor A_λ commutes with M_{22} , for any matrix A_λ on M_λ . In a three-dimensional multiplicity carrier the full $M_3(C)$ algebra is therefore allowed by M_{22} equivariance.

For any desired unitary frame W , the two positive operators D_u and $W D_d W^\dagger$ are both permitted on such a multiplicity space. Their eigenvalues can be distinct. Their relative eigenframe is W . Consequently M_{22} equivariance alone accommodates every CKM frame and cannot force the golden relation or a particular CP phase. Canonical orbital geometry supplies distinguished candidate operators, but selecting specific orbital coefficients requires an additional interaction or residual symmetry. Algebra-generation and transport-isomorphism theorems do not perform that selection.

This is a scoped obstruction to using the unbroken commutant as the complete flavor principle. It is not a no-go theorem for M_{22} breaking, decorated incidence fields, projective carriers, or a more restrictive interaction category.

The canonical C3/C7 frame fails

The first frame test takes the up-sector Hermitian operator to be X and the down-sector operator to be iY . Positive shifts, positive overall rescalings, and scalar functions that preserve the nondegenerate eigenprojectors do not change this frame test. Both ordinary real three-dimensional carriers have simple spectra for X and iY .

On the first real carrier the X eigenvalues are $-5, 3, 30$ and the iY eigenvalues are $-\sqrt{1452}, 0, \sqrt{1452}$. Its squared overlap matrix, in increasing spectral order, has rows approximately $(0.422078, 0.155844, 0.422078)$, $(0.196970, 0.606061, 0.196970)$, and $(0.380952, 0.238095, 0.380952)$. On the second real carrier X has eigenvalues $-5, -2, 3$ and iY has eigenvalues $-\sqrt{2732}, 0, \sqrt{2732}$. Its rows are approximately $(0.00512445, 0.98975110, 0.00512445)$, $(0.49897511, 0.00204978, 0.49897511)$, and $(0.49590044, 0.00819912, 0.49590044)$.

The equality of the outer probability columns is structural. In a real carrier X has a real eigenbasis, while a real skew Y has conjugate eigenvectors for the opposite eigenvalues of iY . Their squared overlaps with any real X eigenvector are equal. The numerical column differences are below $6\text{e-}16$. Row and column permutations retain the identical-column obstruction. Two identical probability columns cannot each concentrate near one in different rows, so neither frame has the CKM hierarchy. This statement does not require interpreting a convention-dependent eigenvector phase as the CKM phase.

The other two three-dimensional carriers are a conjugate complex pair, so the preceding real-carrier argument must not be applied to them individually. Both have X spectrum $-2, -2, 3$. The repeated eigenvalue leaves a $U(2)$ freedom and prevents X from selecting a unique family frame. The unique X eigendirection has overlaps approximately $0.0381554, 0.5982882, 0.3635564$ with the iY eigenlines, in reverse order on the conjugate carrier. These overlaps are unchanged by rotations in the repeated eigenspace. Their maximum

is below 0.599, so even that fixed direction cannot be a nearly unmixed CKM row under any permutation. The receipt omits a full mixing matrix for these degenerate cases because it would depend on arbitrary eigenbasis choices.

Thus all four available three-dimensional blocks fail for this canonical pair. This does not exclude pairs built from additional unsigned channels, higher words, decorated C5 incidence, or another vacuum-selected Hermitian operator.

The twelvefold cover supplies a phase carrier

The primary ATLAS data gives M22 Schur multiplier of order twelve and lists its twelvefold cover. The recovered Formation work already tracks separate 2-primary and 3-primary cover restrictions. The full universal cover has central C12. An order-five element of M22 can be lifted to an order-five element of the cover: its fifth power lies in C12, and the fifth-power map is invertible on C12, so a central correction cancels that power. This lift commutes with the center. Their subgroup is C12 times C5, hence C60.

This is a group-theoretic inference from the cover data. It avoids the ordinary-M22 restriction that a single group element has no order sixty. It is not a claim that the old untwisted 7,392-state permutation module sees the central character; that module factors through M22 and its center acts trivially.

A character of this C60 subgroup has phase residue $n=5a+12b$ modulo 60, with a modulo 12 and b modulo 5. All sixty residues occur. The primitive positive phases are 6,42,66,78,102,114,138,174 degrees. The residue eleven, corresponding to 66 degrees, can be written with $a=7$ and $b=3$. These are possible character labels, not dynamically selected quantum numbers.

If one retains the previously nominated readout $C=1-2 \cos(12 \theta)$, the primitive residues n congruent to plus or minus one modulo five give $C=(3-\sqrt{5})/2=\phi^{-2}$. Its exact polynomial is C^2-3C+1 . The positive angles on this branch are 6,66,114,174 degrees. Thus the cover supplies both the relevant cyclotomic arithmetic and several competing branches. It neither selects 66 degrees nor derives the equality of this singlet readout with the observable CKM coefficient. The other primitive residues give the larger golden root.

Central phases do not select a residual eigenframe. Multiplying a residual group matrix by a central scalar changes its eigenvalues but leaves all its spectral projectors unchanged. Such scalar factors therefore cancel from projector overlaps. Similarly an overall Yukawa or Yukawa-column phase cancels from YY-dagger. A primitive eigenphase of 66 degrees is not evidence for a physical CKM phase of 66 degrees. Multiple charge-compatible interfering contractions and a nonzero rephasing invariant must be demonstrated explicitly.

The next interaction calculation

The useful continuation is a decorated order-five incidence construction coupled to the recovered signed C7 channels. First derive the allowed cover characters and local contractions from the actual Schur restrictions. A faithful central character cannot simply be attached to an ordinary A5 triplet without checking its extension and charge constraints. Build the resulting twisted or decorated carrier and identify two Hermitian operators with simple three-dimensional eigenframes. Their coefficients must follow from the permitted contractions or a solved vacuum, rather than from a fitted CKM matrix.

Then compute the relative spectral-projector overlaps and a CP-sensitive closed path invariant. Check whether any central phase survives these neutral observables. Fit only the existing two anchor magnitudes, if the construction admits two independently justified scales; evaluate the remaining CKM magnitude, effective golden coefficient, and physical phase as predictions. Enumerate every allowed orientation and character branch, including CP conjugates and the four golden-angle candidates. Finally enumerate symmetry-allowed counterterms to test whether the frame survives coefficient changes.

The recovered algebra has enough structure to make this calculation concrete. What it currently establishes is an exact operator reconstruction, a failure of the simplest canonical frame, and an order-60 cover route with an explicit branch and rephasing obstruction. A symmetry-protected quark frame has not been derived.

Reproduction and sources

Run `python python/develop_m22_frame.py` from the repository root with NumPy, SciPy, SymPy and python-flint installed. The output operator matrices and transpose/subdegree data are in `receipts/m22_frame/recovered_operators.json`. The diagnostic, exact polynomial factors and phase enumeration are in `frame_audit.json`. The trusted original object-array archive is the same one used by the byte-preserved Forge verifier; do not replace it with an untrusted NumPy pickle.

Recovered source theorems include `RAMANUJAN_WILSON_FAMILY.md`, `WEDDERBURN_35_RAMANUJAN_REDUCTION.md`, `RAW_FIVE_BLOCK_STAR_TRANSPORT_v11.md`, `CHAR0_TRANSPORT_GLUE_M5_v13.md`, `M22_PRIMARY_SCHUR_SIGNATURES_v11.md` and `L34_MATCHED_SELECTOR_V24.md` under `galois_merge/vendor/forge/THEOREMS`. The primary external cover reference is the ATLAS M22 page, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M22/>. Its multiplier, cover and conjugacy-class data were checked on 5 October 2026. The CKM target and the mediator matching remain those of the existing protected-flavor chapters.

Triplet transport and exact residual obstructions

Research manuscript and continuation of the protected-flavor monograph. Prepared 5 October 2026. Status: computationally certified mathematical results with proofs; publication novelty and the proposed particle-physics interpretation require independent review. No golden CKM coefficient, CKM phase, or experimental mixing anchor enters the group, connection, or overlap enumeration.

Abstract and the claim being tested

We reconstruct a canonical decorated connection on a 7,392-cap M22/A5 geometry recovered from the Wilson work. Its order-five scalar adjacency B satisfies $B^2 = X + 6I$ in the preserved 152-dimensional commutant, so it cannot supply eigenframes independent of X . The cap graph consists of 176 distance-regular components with 42 vertices, with intersection array $\{6,5,1;1,1,6\}$. Each cap has five A4 subgroups. Matching them across a common D5 by their common involutions defines a parameter-free five-letter edge permutation. The resulting holonomy is S5. On the exterior square of the standard four-dimensional hyperplane representation, every edge exchanges the two local golden A5 triplets. There is no parallel rank-three projector on the base graph. Its connected 84-vertex orientation cover has A5 holonomy and admits exactly two such projectors; a positive interaction selects them. Independently, an exact census of the genuine complex triplet of 3.A6 gives 151 complete Abelian residual frames and 111 squared entry values. Every nonzero entry has magnitude at least 0.0759431758699. Two involution residuals also have an exact fixed-entry obstruction. These results identify concrete failures and a precise alignment repair, rather than deriving the nominated golden CKM relation.

The central question is whether the recovered symmetry can enforce $C_{\text{eff}} = |\text{Vub}|(1-|\text{Vub}|^2)/(|\text{Vus}||\text{Vcb}|) = \phi^{(-2)}$ and a physical CKM phase of 66 degrees. The results below do not establish either equality. In particular, geometric transport, a representation's character field, a central scalar phase, and the physical CKM invariant are different objects.

Exact scalar identity and the 42-cap component

$$B^2 = X + 6I, \quad B^3 = 4X + 11B + 6D$$

The trusted source is `galois_merge/vendor/forge/DATA/RELATION/TASK_SELECTED_WORD_PROGRAMS_V23/exact_prefix_1023.npz`, SHA256

1351e514fcb5f23832f65665e0a12d2e172f8131c34875de0e03cddc5aec3093. It records the original integral Hecke coordinates of 1, X, Y and their binary words. The original X is orbital A49, and the original Y is -A12-A48-A59+A95+A116+A125. The sole valency-six orbital is B = A134; the sole valency-five orbital is D = A139. Solving against 152 independent preserved words and checking all 511 available extensions reconstructs multiplication by B and D without CKM input.

Theorem 1. In the full reconstructed regular commutant representation, $B^2 = X + 6I$ and $B^3 = 4X + 11B + 6D$. Consequently $B^4 - 4B^3 - 17B^2 + 24B + 36I = 0$. Its regular characteristic polynomial is $(t-6)^9(t-2)^{65}(t+1)^{27}(t+3)^{51}$. These are exact integral matrix identities, not rounded

spectral matches.

Proof. Express the orbitals B and D in the selected exact word basis using rational matrix inversion. Their recovered multiplication matrices have integral entries. Direct integer matrix products give the stated identities. The characteristic polynomial is computed independently from the integral matrix. The complete matrices and replay script are supplied with the receipts. The four distinct roots occur, so the displayed quartic is the minimal polynomial.

An independent degree-22 permutation construction uses the ATLAS standard M22 generators. A subgroup generated by an involution and an element of order three with product of order five has order 60. The selected cap class has subdegree counts $1^1, 5^1, 6^1, 10^1, 15^2, 20^4, 30^{42}, 60^{100}$, matching the recovered action. Enumerating its cosets accounts for all 443,520 group elements and 7,392 caps. Each cap contains six C5 subgroups; a C5 lies in exactly two caps in this class. The corresponding undirected cap graph has valency six. The implementation finds a 42-vertex component with distance layers 1,6,30,5 and the stated intersection array; transitivity gives 176 components.

The component graph is a known Hoffman-Singleton second-subconstituent graph. Its spectrum is $6^1, 2^{21}, (-1)^6, (-3)^{14}$. Neither the graph nor its intersection array is claimed as a new discovery. The five distance-three neighbors together with a vertex form an antipodal six-cap block; the component has seven such blocks. The exact component stabilizer acts faithfully as all 2,520 even permutations of the seven blocks, identifying it as A7. A six-cap block stabilizer acts faithfully as all 360 even permutations of its six caps, identifying an ordinary A6 containing the cap A5 with index six.

The identity $B^2 = X + 6I$ means every X eigenspace is B-invariant and conversely B selects only branches already lying in these spaces. Since the recovered X eigenvalues correspond to four distinct B^2 values, B and X have the same eigenspace decomposition. The order-five scalar operator therefore does not repair the earlier failure to obtain a new three-family mixing frame.

A canonical decorated transport

$$\Omega = \frac{EG_2}{\sqrt{5}}, \quad \Omega^2 = I, \quad \Omega T = -T\Omega$$

Each A5 cap has five A4 subgroups, equivalently the normalizers of its five Klein four subgroups. Adjacent caps intersect in D5. Each local A4 meets this D5 in a single order-two subgroup, and the five A4 subgroups exhaust its five involutions. Match the source and target A4 subgroups with the same actual involution. This defines a unique bijection of the five local letters, hence a permutation edge transport. The construction is covariant under M22 conjugation and has no continuous coupling or chosen CKM angle.

Theorem 2. For this incidence-matching connection, every edge permutation is odd and the holonomy of a 42-cap component is S5. On $W = \text{exterior-square } R$, where R is the four-dimensional standard hyperplane representation of the local five letters, transport exchanges the self-dual and anti-self-dual three-dimensional subspaces.

Proof. The supplied cap permutations identify all local subgroups and all 252 directed edge transports by exact set intersections. Every edge is odd. Products around loops, after a spanning-tree trivialization, generate all 120 permutations of five letters. An independently written verifier recomputes every transport from conjugated A4 and cap intersections, uses a depth-first rather than breadth-first tree, and again obtains the entire S5. One explicit five-cycle is cap 0, cap 1083, cap 2333, cap 3074, cap 5702, cap 0. Its loop permutation is (2,0,4,3,1) in zero-based image notation, an odd permutation. Vertex labels refer to the supplied deterministic

coset enumeration.

For the representation-theoretic statement, use the hyperplane basis $e_i - e_5$, $i=1, \dots, 4$, with metric $G = I + 11^T$, of determinant five. Let G_2 be its induced exterior-square metric and E the matrix of wedge pairing in the six ordered bivector coordinates. The Hodge operator is $\Omega = E G_2 / \sqrt{5}$, and $\Omega^2 = I$. Its two projectors $(I \pm \Omega)/2$ have rank three. For any five-letter permutation p , its induced bivector transport T satisfies $\Omega T = \text{sign}(p) T \Omega$. Every edge is odd, hence maps one Hodge branch to the other. The local A_5 preserves the branches. Their order-five characters are $(1+\sqrt{5})/2$ and $(1-\sqrt{5})/2$ in opposite classes, identifying the familiar two real golden triplets. This is a real representation-theoretic branch exchange, not a CP phase calculation.

For an independent irreducibility check, the bivector character is $\chi(p) = ((\text{fix}(p)-1)^2 - (\text{fix}(p^2)-1))/2$. Its squared character sum is 120 over S_5 , giving inner product one, and is 120 over A_5 , giving inner product two. The full S_5 commutant is therefore scalar, while the A_5 commutant is spanned by I and Ω . Exact commutator-rank calculations reproduce dimensions one and two.

Corollary. There is no parallel Hermitian rank-three projector on this connected base connection. Any such projector would commute with every loop at the root. The irreducible S_5 action forces it to be a scalar idempotent, hence either zero or identity, of rank zero or six. The theorem excludes parallel projectors for the specified connection; it does not prohibit homogeneous A_5 triplet bundles or all possible M_{22} models.

What edge symmetry actually enforces

$$\dim \text{Hom}_{D_5}(3, 3^t) = 1, \quad \dim \text{Hom}_{D_5}(3, 3^{\bar{t}}) = 2$$

The D_5 intersection reveals an additional coefficient issue. The restriction of one golden triplet to D_5 is a one-dimensional sign axis plus a two-dimensional plane carrying C_5 weights ± 1 ; its Galois partner has the same axis and plane weights ± 2 . Transport between adjacent caps exchanges the two five-cycle character classes. Taking the exact character inner products over the ten common elements gives dimension one for a same-branch edge intertwiner and dimension two for an opposite-branch edge intertwiner.

Thus a same-branch covariant map has rank at most one. An opposite-branch map can be invertible, but its axis and plane couplings are independent. The particular incidence matching above picks a unitary map on both channels; D_5 symmetry alone does not force their relative strengths. It would be incorrect to turn this particular geometric isometry into a general coupling-protection theorem. A particle interaction must explain why it uses the incidence matching rather than the full symmetry-allowed two-channel family.

Multiplying edges by central scalar phases, including phases from a central cover, leaves $\Omega T = -T \Omega$ unchanged. A phase factor also cancels in $T P T^{-1}$, so it cannot repair the parallel-projector obstruction. A noncentral transport or different fiber content could do so; neither is excluded here. No identification of a cover eigenphase with the CKM phase is made.

An explicit protected alignment interaction

$$J_{V,s} = (-1)^s \Omega_V, \quad P_{\pm} = \frac{I \pm J}{2}$$

The orientation cover has vertices (v,s) , with $s=0,1$, and lifts an edge $v-w$ to $(v,s)-(w,1-s)$. It is connected because the base has odd loops. Each component has 84 vertices. The lifted

holonomy consists exactly of the 60 even permutations, A_5 . The field $J_{(v,s)} = (-1)^s \Omega_v$ is parallel because edge transport anticommutes with Ω . The deck exchange changes the sign of J .

Theorem 3. On this cover the only parallel Hermitian rank-three projectors are $P_+ = (I+J)/2$ and $P_- = (I-J)/2$. For dimensionless Hermitian endomorphism order parameters P_a , an explicit potential selecting exactly these configurations is the sum of three nonnegative terms: a positive coefficient λ times the sum of squared norms of $P_a^2 - P_a$, a positive coefficient μ times the sum of $(\text{Tr } P_a - 3)^2$, and a positive coefficient κ times the sum over edges of squared norms of $P_b - T_{ba} P_a T_{ba}^{-1}$. The norms use the positive fiber metric.

Proof. Zero energy enforces a rank-three projector and parallel transport. At a root the A_5 commutant is $\text{span}\{I, J\}$. A Hermitian idempotent in this commutant has eigenvalues $a+b$ and $a-b$ on the two inequivalent triplets. Rank three forces these to be 1 and 0 in either order. This gives exactly P_+ and P_- . Conversely, their parallelism and rank make all terms vanish. On the base the identical constraints have no zero-energy configuration, by the scalar S_5 commutant. The cover is therefore a concrete repair of this alignment problem, rather than a choice of a fitted frame.

The selected projectors remain exact for arbitrary positive λ , μ and κ within this stated interaction. This is a classical robustness result for the declared potential family. Symmetry allows additional interactions; no general protection against counterterms or radiative corrections is claimed.

This establishes a local rank-three subspace. It does not by itself yield exactly three light global quark modes, nondegenerate quark masses, a relative up/down frame, a UV origin for the interaction, or a cosmological selection of its two branches. In particular, the two real Hodge sectors are not the positive and negative physical CP vacua. Additional physical fields and a light-mode analysis are necessary before treating the construction as a flavor model.

Exact complex-triplet residual gap

$$|V_{ij}| \in \{0\} \cup [0.0759431758\dots, 0.9566183482\dots] \cup \{1\}$$

The identified A_6 block supplies a concrete place to investigate a genuine complex triplet. Ordinary A_6 has no faithful ordinary triplet; its triple cover $3.A_6$, the Valentiner group $\text{Sigma}(360 \times 3)$, does. We use the published generators of Hagedorn, Meroni and Vitale, equations (7) and (33), and close them exactly in $Q(\sqrt{5}, \omega)$, with $\omega^2 + \omega + 1 = 0$. The closure has 1,080 elements and a center of order three, and its real A_5 subgroup has order 60. A consistent embedding of this triple cover into a particular central M_{22} cover is a separate requirement, not an assumption silently imposed by this calculation.

The complete minimal Abelian residual frames comprise 40 order-three frames, 45 order-four frames, 36 order-five frames, and 30 joint Klein frames, totaling 151. The order-three frame count is smaller than the number of cyclic order-three subgroups because central multiplication gives the same projectors. Exact spectral-projector enumeration checks coverage of all 474 noncentral elements of orders three, four and five. Order-twelve and order-fifteen lifts have the same frames as their cubes; order-two and order-six residuals leave a two-dimensional degeneracy. Higher Abelian residuals add the center or already contain one of the complete frames. This classification agrees with the published subgroup analysis; it is not presented as a newly discovered flavor group.

Theorem 4. Every entry probability between the 151 complete frames belongs to the supplied exact set of 111 values. The smallest positive value is $p_{\min} = 0.005767365961210174\dots$, and the largest value strictly below one is $p_{\max} = 0.9151186641666433\dots$. Consequently every nonzero magnitude is at least $0.07594317586992\dots$, and every subunit magnitude is at most $0.956618348228092\dots$. Rephasing and all row and column mass permutations preserve these bounds.

Proof. Construct cyclic projectors as the finite Fourier sums of exact group powers, retaining the three projectors of trace one. Construct Klein projectors by their four characters, retaining the three projectors of trace one. For all 11,476 unordered frame pairs with repetition, evaluate all nine traces $\text{Tr}(P_i Q_j)$, giving 103,284 exact comparisons. Computation is in $\mathbb{Q}(\zeta_{60})$, reduced modulo Φ_{60} . All probabilities are exactly fixed by complex conjugation. Convert them to degree-seven rational polynomials in $t = \zeta_{60} + \zeta_{60}^{-1}$. Its minimal polynomial is $t^8 - 7t^6 + 14t^4 - 8t^2 + 1$, and the physical embedding is its largest real root, $2 \cos(\pi/30)$. Rational root isolation and interval Horner evaluation give strictly ordered, disjoint rational enclosures for all 111 values. This proves both extrema and their ordering without using floating point to choose the winner. The 60th root is a common arithmetic field for orders 3,4,5; its appearance is not a prediction of 66 degrees.

Explicitly, $p_{\min} = -t^{7/10} - t^{6/20} + 11t^{5/20} + 3t^{4/10} - 3t^{3/5} - 9t^2/20 - t/5 + 1/2$. The upper endpoint below one is $p_{\max} = 2t^{6/9} - 4t^{4/3} + 2t^2 + 1/9$. The full rational polynomial table and certified intervals are in `receipts/m22_interactions/valentiner_exact_gap.json`. A separate unitary numerical census of all 22,801 ordered frame pairs independently reproduces the extrema; 36 mass label permutations per pair correspond to 820,836 labeled pairs, and no labeling changes the entry spectrum.

The nominated values $|V_{cb}|=0.0411$ and $|V_{ub}|=0.00352144218276$ fall inside the forbidden nonzero gap. Thus complete Abelian residual symmetries cannot give that CKM hierarchy in this triplet without symmetry breaking. These numbers are used only after the exact spectrum is constructed, as a comparison to the existing nominated model, not as a current experimental global fit. The result excludes the specified exact residual construction; it does not exclude 3.A6 as a broken flavor symmetry.

Partial residuals and the fixed-line obstruction

$$|V_{ij}|^2 \in \left\{ 0, \frac{3-\sqrt{5}}{8}, \frac{1}{4}, \frac{1}{2}, \frac{3+\sqrt{5}}{8}, 1 \right\}$$

Leaving a two-dimensional degenerate sector in each quark mass matrix permits continuous rotations, so the preceding complete-frame gap does not apply directly. Nevertheless, retaining an involution in both sectors fixes one line in each. There are 45 order-two elements. Their rank-one projectors are $(I+g)/2$. Evaluating all 2,025 ordered projector overlaps exactly gives squared fixed entries $0, (3-\sqrt{5})/8, 1/4, 1/2, (3+\sqrt{5})/8$, and 1 . Their magnitudes are $0, 0.309016994\dots, 0.5, 0.707106781\dots, 0.809016994\dots$, and 1 .

Theorem 5. If both nondegenerate quark Hermitian mass matrices commute with nontrivial involutions in the stated triplet, at least one CKM entry is in this fixed set, irrespective of the two otherwise arbitrary $U(2)$ rotations and irrespective of any added compatible CP constraint. Distinct fixed lines cannot give the nominated hierarchical CKM matrix. Identical fixed lines give an exactly decoupled family and hence a two-family mixing block. A zero fixed overlap also supplies an exact zero entry. All these cases require symmetry breaking to realize the nominated matrix with nine nonzero entries.

Proof. Commutation with an involution preserves its one-dimensional $+1$ eigenspace and its two-dimensional -1 eigenspace. Nondegenerate mass eigenstates therefore include the fixed line in each sector. Their overlap is a CKM entry. Rotations confined to the two-dimensional complements do not change this overlap. The exact projector calculation gives the stated list. The nominated matrix's off-diagonal magnitudes are below 0.3090 and its diagonal magnitudes exceed 0.8091; none is zero or one. Adding CP cannot change the representation-fixed line overlap. This argument does not assume a specific CP automorphism or a chosen mixing phase.

Publication scope and the next physical problem

The result suitable for scrutiny is a conjunction of exact transport obstruction, explicit cover alignment mechanism, and exact quark residual gaps. It is not a derivation of the golden CKM relation. Known ingredients include the graph, the two A5 triplets, the Valentiner generators and residual-subgroup classification, and the Hodge transformation law. The documented contribution is their explicit reconstruction on the recovered cap system, the canonical involution-matching connection with certified holonomy, the scoped interaction repair, and exact gap certificates applied to the nominated quark mechanism. A literature search did not establish novelty of this combination; independent specialist review remains necessary before submitting a novelty claim.

The remaining viable problem is now narrower. A proposed theory must specify a noncentral, charge-compatible interaction that produces three light quark modes, breaks the exact residual obstruction in a controlled way, and correlates the up/down breaking operators after canonical normalization. It must then derive the coefficient $\phi^{(-2)}$ and a neutral CP observable corresponding to 66 degrees without inserting either target. Central phases alone, scalar order-five adjacency alone, and exact complete or two-involution residual frames cannot supply that result. Small free corrections that merely fit the missing entries would not resolve the original coupling-protection problem.

Reproduction and sources

Install `python/m22_frame_requirements.txt`. From the repository root run `python python/certify_m22_c5.py`, `python python/develop_m22_transport.py`, `python python/verify_m22_transport.py`, `python python/develop_valentiner_frames.py`, and `python python/certify_valentiner_gap.py`. The first three scripts distinguish the preserved Wilson regular algebra from the independently enumerated cap geometry. The independent verifier reconstructs subgroup intersections directly and does not reuse the connection's cocycle code or centralizer solver. The exact residual-gap script checks its entire field and real embedding using rational intervals. Logs and JSON receipts are included. A cold replay compares JSON outputs byte for byte; the compressed permutation file is validated by its array contents, not by ZIP container metadata. No CKM input enters any of these five computations.

The standard degree-22 generators are from the ATLAS of Finite Group Representations, <https://brauer.maths.qmul.ac.uk/Atlas/spor/M22/>, generator files `gap/M22G1-p22B0.g1` and `gap/M22G1-p22B0.g2`. Thomas Connor and Dimitri Leemans, The subgroup lattice of M22 (2012), <https://leemans.dimitri.web.ulb.be/atlaslat/m22.pdf>, independently documents the relevant A5, A6 and A7 subgroup classes; the selected class is recorded by explicit permutations rather than inferred only from a class name. Claudia Hagedorn, Aurora Meroni and Lorenzo Vitale, Mixing Patterns from the Groups $\Sigma(n, \phi)$, arXiv:1307.5308, <https://arxiv.org/abs/1307.5308>, supplies the Valentiner generators and prior residual analysis. The known 42-vertex graph is described in Wei Jin and Cheryl E. Praeger, Normal quotients of

diameter at most two of finite three-geodesic-transitive graphs, *Journal of Combinatorial Theory A* 178 (2021), 105349, <https://doi.org/10.1016/j.jcta.2020.105349>. These sources establish antecedents; they do not independently verify the new receipts or confer a publication claim.

Valentiner interactions and the protection boundary

Result and scope

The recovered cap geometry has a supported central-cover route to a genuine complex triplet. The triplet's first new symmetric invariant is a unique sextic, derived here directly from its exact generators. A minimal vectorlike mediator model has exactly three light modes per charge sector before electroweak breaking, and its canonical matching transmits the flavon frame. Its complete scalar operator space nevertheless contains independent cross-sector quartics. A target-free numerical local-vacuum experiment shows one such operator changing the observable CKM coefficient. Neither the golden coefficient nor the 66-degree physical phase is derived.

These results replace an unspecified symmetry search with a concrete representation, a fixed invariant tensor, specified field content and a counterterm audit. The sextic itself is the classical Valentiner/Wiman invariant, not a newly discovered polynomial. Publication novelty of the combined construction and obstruction requires comparison with the existing invariant-theory and flavor literature.

The central-cover connection

The earlier cap computation established a component stabilizer A_7 and a six-cap block stabilizer A_6 of index seven in that A_7 . Both A_7 conjugacy classes in M_{22} lift to $3.A_7$ in $3.M_{22}$. The GAP Character Table Library lists the index-seven maximal subgroup of $3.A_7$ as $3.A_6$, of order 1080. Thus the full preimage of the cap-block A_6 is the nonsplit Valentiner cover, rather than the split product $C_3 \times A_6$. This establishes the abstract central-cover chain $3.A_6 < 3.A_7 < 3.M_{22}$ for the recovered stabilizer geometry.

This conclusion uses the independently computed ordinary stabilizers together with published subgroup classifications. We have not constructed an explicit matrix representation of $3.M_{22}$ or lifted every recovered permutation generator. The local triplet does not extend to a three-dimensional representation of the whole cover by this argument. A global bundle, a global mass operator on 7392 caps, and decoupling of its other modes remain separate obligations.

Primary sources:

<https://www.math.rwth-aachen.de/~Thomas.Breuer/ctbllib/ctbltoc/data/3.M22.html> ;

<https://www.math.rwth-aachen.de/~Thomas.Breuer/ctbllib/ctbltoc/data/3.A7.html> ;

<https://www.math.rwth-aachen.de/homes/sam/ctbllib/ctbltoc/data/3.A6.html> . The exact local $SU(3)$ generators are those of Hagedorn, Meroni and Vitale, <https://arxiv.org/abs/1307.5308> , equations (7,33).

An exact tensor boundary

Let R be the faithful complex triplet and $G=3.A_6$. We closed its four generators exactly in $Q(\sqrt{5}, \omega)$, with ω squared plus ω plus one equal to zero, obtaining all 1080 elements. For every pair p, q with total degree at most eight, we evaluated the character average $\dim \text{Inv}(R \text{ tensor } p \text{ times } \bar{R} \text{ tensor } q) = \sum_g \text{tr}(R(g))^p \text{conjugate}(\text{tr}(R(g)))^q / 1080$. We independently computed the corresponding $SU(3)$ tensor multiplicities from partitions and hook lengths.

Theorem. At total degree at most five the G -invariant tensor space equals the $SU(3)$ -invariant tensor space, for arbitrarily labeled triplet fields. At degree six, the $(3,3)$ space still has dimension six for both groups. The $(6,0)$ space has dimension six for G and five for $SU(3)$, leaving exactly one new tensor. The conjugate statement holds for $(0,6)$; the other degree-six bidegrees add none.

Proof. $SU(3)$ invariants form a subspace of G invariants. The exact character averages and independent hook-length counts coincide in the stated bidegrees. Equality of dimensions therefore proves equality of spaces. In degree six, the symmetric-power average gives $\dim \text{Inv}(\text{Sym}^6 R) = 1$. Its tensor cannot be an $SU(3)$ symmetric invariant, because there is no positive-degree $SU(3)$ -invariant polynomial on one fundamental triplet. It exhausts the one-dimensional quotient. All character coefficients cancel exactly in the four-dimensional number field. The full counts are in `receipts/m22_interactions/valentiner_invariants.json`.

For one commuting triplet, through degree six the only nonconstant invariant bidegrees are $(1,1)$, $(2,2)$, $(3,3)$, $(6,0)$ and $(0,6)$, each with dimension one. The first three are powers of the norm. Consequently every renormalizable single-triplet potential has an accidental continuous $SU(3)$ orientation symmetry. A sextic is the first possible finite-group anisotropy. This is an effective interaction in four spacetime dimensions, requiring a suppression scale or a specified mediator completion.

The unique sextic in the recovered basis

$$F_6 = x^6 + y^6 + z^6 + A(x^4 y^2 + y^4 z^2 + z^4 x^2) + B(x^2 y^4 + y^2 z^4 + z^2 x^4) + D x^2 y^2 z^2$$

Writing the triplet as (x, y, z) , normalize the sextic by its x^6 coefficient. It is $F_6 = x^6 + y^6 + z^6 + A(x^4 y^2 + y^4 z^2 + z^4 x^2) + B(x^2 y^4 + y^2 z^4 + z^2 x^4) + D x^2 y^2 z^2$, with $A = [15 - 9\sqrt{5} + i(15\sqrt{3} + 3\sqrt{15})]/8$, $B = [15 + 9\sqrt{5} + i(-15\sqrt{3} + 3\sqrt{15})]/8$ and $D = 15 - 3i\sqrt{15}$.

The coefficients arise from solving invariant linear equations, without a selected angle or golden CKM coefficient. Sparse monomial generators first reduce the degree-six space; the remaining generator leaves a single invariant. Each of the four polynomial identities $F_6(g, \phi) = F_6(\phi)$ is then checked exactly in the number field. The receipt records every monomial coefficient.

A potential contains a free complex overall coefficient multiplying F_6 plus its conjugate. Uniqueness fixes internal component ratios; it does not fix the coefficient relative to other invariants. Its complex coefficients also do not establish physical CP violation. A consistent generalized CP action on all fields and its allowed couplings must still be constructed. Ordinary componentwise conjugation is not assumed to be a symmetry in this basis.

Specified quarks, flavons and mediator interactions

Take Q_L in R , the Higgs in the trivial family representation, three right-handed up quarks and three right-handed down quarks in family singlets, and one vectorlike quark triplet F_u and one F_d in R . These vectorlike fields are Standard Model weak singlets with the respective up or down electric charge. Introduce six Standard Model singlet flavons ϕ_i in R . This is a local four-dimensional effective model after selecting the cap stabilizer; it is not a derived global cap spectrum.

Add a specified shaping group C_6 to the sixth power. Each right-handed singlet f_R has charge +1 under its own factor, its flavon has charge -1, and all other listed fields are neutral under that factor. The family and shaping symmetries permit, per sector, $M_f F_{\bar{L}} F_R$, $h_f Q_{\bar{L}} H_f F_R$, and $\sum_i \lambda_i F_{\bar{L}} \phi_i f_R$, plus Hermitian conjugates. Here H_u is the conjugate Higgs doublet and H_d the ordinary one. Each family contraction is unique, by $\dim \text{Hom}_G(R, R) = 1$. The mediator mass is therefore flavor-universal. Cross-column flavon couplings are forbidden by the specified shaping charges. A bare $Q_{\bar{L}} H_f f_R$ coupling has no family singlet. A mediator-triplet bilinear with one triplet flavon is forbidden by the nontrivial central charge. These exhaust renormalizable mass and Yukawa structures for this listed field content.

The symmetry allows separate M_f , h_f and three λ_i in each sector. It does not correlate the up and down parameters. The shaping group is an explicit model assumption, not something derived from M_{22} . Its gauging, anomalies, a scalar messenger completion and quantum thresholds are not claimed.

Three light modes and canonical projectors

$$K_f = I + A_f^\dagger A_f, \quad Y_f = -h_f A_f K_f^{-1/2}, \quad |V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$$

At $H=0$ define C_f with columns λ_i times the flavon expectation values. The heavy row block is $[C_f, M_f I]$, with three left-handed rows and six right-handed columns. For nonzero M_f it has rank exactly three, regardless of C_f . Its right kernel therefore has dimension three; the three Q_L components are the corresponding left chiral light modes. The three heavy singular values are $\sqrt{|M_f|^2 + \sum_i \lambda_i^2}$. There are exactly three light families per charge sector in this specified local model, without interpreting a rank-three fiber as a global family count.

Set $A_f = C_f / M_f$, $K_f = I + A_f^\dagger A_f$. The normalized light right frame is $[I; -A_f] K_f$ to the power $-1/2$, and the matched Yukawa matrix is $Y_f = -h_f A_f K_f$ to the power $-1/2$. These are exact tree identities at $H=0$, with no small-flavon expansion. They imply $H_f = Y_f Y_f^\dagger = |h_f|^2 S_f (|M_f|^2 I + S_f)$ inverse, where $S_f = C_f C_f^\dagger$.

The positive spectral function is strictly increasing. Thus the left spectral projectors of H_f and S_f agree, including the mass ordering for a nondegenerate spectrum. Quark mixing is computed as $|V_{ij}|^2 = \text{Tr}(P_i^u P_j^d)$, and $J = \text{Im Tr}(P_1^u P_1^d P_2^u P_2^d)$. Arbitrary flavon frames remain arbitrary physical quark frames; canonical normalization neither creates nor protects the golden relation. At finite Higgs expectation value, heavy-light mixing supplies higher-dimensional EFT corrections and potential charged-current nonunitarity. Those corrections are not included in this matching statement.

The complete scalar space through degree six

For the six specified flavons and C6 shaping factors, a degree smaller than six has equal numbers of each flavon and its conjugate. Consequently the scalar operators coincide with balanced Gram contractions: six quadratic norms; 36 quartics consisting of 21 norm products and 15 absolute overlap squares. With one Higgs, add its quadratic norm, its quartic norm and six Higgs-norm/flavon-norm products, for 50 real renormalizable scalar couplings in total.

At degree six the general Hermitian flavon space has 198 real parameters: 56 cubic norm products, 90 norm-times-overlap squares, the real and imaginary parts of 20 Gram triangles, and the real and imaginary parts of six individual $F6(\phi_i)$. Balanced epsilon contractions are Gram determinants and add no independent operators. Gram rank relations first occur at degree eight. The shaping charges forbid mixed polarizations of the new sextic at degree six: an unbalanced field must occur six times by itself. Under a specified CP action the count can decrease, but no CP classification has been assumed here. Every absolute overlap square remains CP-even under a common unitary anti-linear CP action on the triplets.

In particular, $\epsilon |\phi_{u1}^\dagger \phi_{d1}|^2$ squared is an independent allowed quartic. Neither the group nor the shaping charges constrain its coefficient to the chosen sextic coefficient. This is a concrete obstruction to claiming protected alignment merely from uniqueness of F6. It is the interaction-level counterpart of the earlier allowed linear angular deformation, although no equality between those two coordinate descriptions is assumed.

A target-free local-vacuum deformation experiment

We minimized the phase of each sextic analytically, fixed each flavon norm to one, and considered the reduced potential $\sum_i -|F6(\phi_i)| + \epsilon |\phi_{u1}^\dagger \phi_{d1}|^2$ squared. This is the fixed-norm reduction of polynomial sextic terms plus their conjugates, with independent phases minimized. Radial stability requires separate radial terms; no global minimum or cosmological vacuum selection is certified.

Six independently generated sextic stationary directions were selected from the numerical group orbit, and their column strengths set to 0.07, 0.31 and 0.9 in each sector. None is a measured CKM anchor or the golden target. Analytic gradients are checked against independent central finite differences, and each numerical branch must have gradient norm below $1e-7$. The complete matrices, CP invariant and effective golden coefficient are recorded in `receipts/m22_interactions/valentiner_alignment_response.json`. For $\epsilon=0$, C_{eff} is 0.4830422217 and J is 0.01663167860. At $\epsilon=0.01$ they become 0.4831824152 and 0.01663578380. The intermediate positive and negative 0.001 cases also give distinct values. The $\epsilon=0$ and $\epsilon=0.01$ branches have different C_{eff} ; this change survives the exact canonical mediator matching theorem. The mixing is not phenomenologically realistic, which is appropriate for an allowed-deformation counterexample rather than a fitted proposal.

The found single-flavon extrema form a numerical 60-line orbit, and sixteen independent searches reach modulus approximately 1.0886621079. These are numerical observations, not an exact orbit classification or a proof of the global modulus bound. Optimizer success flags and final gradients are retained, including precision-loss flags when a sufficiently small verified gradient remains.

What the next derivation must add

The sextic supplies a real representation-derived coefficient pattern, and the mediator calculation supplies a concrete light-family and normalization result. Their combination does not protect the observable golden relation. The next justified search must introduce interactions that correlate the sectors while controlling all 15 independent quartic overlaps, permit useful mixed sextic contractions without opening arbitrary singlet-copy matrices, and define a consistent generalized CP action. A proposed mechanism must then demonstrate its chosen spectrum and vacuum, test all remaining free parameters, and match the observable projectors. No choice of the desired CKM angle enters the invariant derivation above.

Reproduction

Run `python python/develop_valentiner_invariants.py`, then `python python/develop_valentiner_alignment.py` from the repository. The first command produces an exact invariant receipt and 24 canonical-matching scenarios. The second produces a numerical local-branch receipt with an independent gradient check. The exact group code and dependencies are shared with `python/develop_valentiner_frames.py` and `python/m22_frame_requirements.txt`. The monograph renderer includes this complete chapter. No new Lean theorem or experimental measurement is asserted.

Product-family CP and the global spectrum

Concrete advance and remaining boundary

This push constrains the problematic cross-sector interaction by changing the specified family group to $G_u \times G_d$, with each $G_f=3A_6$, and joining the sectors with a bifundamental link. The bare absolute overlap quartic is forbidden. The first dressed overlap is degree six. The link has a unique new holomorphic sextic tensor, whose component ratios come from the previously derived Valentiner invariant. A renormalizable supersymmetric messenger chain generates this contraction at leading tree order, with its complete allowed holomorphic operator audit retained. A positive alignment interaction has exact zero-energy states at every group matrix, without inserting a CKM angle.

We also construct an exact anti-linear involution on the specified triplet and singlet fields. It leaves the sextic conjugate as required, and a canonical quark calculation produces a pair with opposite nonzero Jarlskog invariants. This establishes physical CP violation in a conditional matched example; it does not derive a unique phase or select the physical vacuum. Finally, the canonical cap connection is certified to have no zero modes. A fully specified modified global mass-squared operator has exactly three zero modes among 44,352 modes, with a gap of one half. Its required symmetry breaking and its chiral quark completion remain independent assumptions.

Exact CP on the elementary field content

$$\overline{X}R(g)X^{-1} = R(u(g)), \quad X\overline{X} = I$$

Use the published outer automorphism that sends the faithful triplet to its conjugate: a maps to a^2 , h to q , f to f^2 , h to q , and q to a . We evaluate these words on the exact generators, propagate the map through all 1080 elements, and check that every repeated group word has the same image. We solve $X \text{conjugate}(R(g)) = R(u(g)) X$ as exact linear equations over $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$. The published representative can square to an inner action. An explicitly recorded inner adjustment yields a symmetric unitary X with $X \text{conjugate}(X)=I$.

The receipt gives the matrix entries, its normalization and the adjustment word. Before normalization, $X_0^\dagger X_0 = (6-2\sqrt{5})I$. The sextic identity is checked coefficient by coefficient: $F_6(X_0 \phi) = (6-2\sqrt{5})^3 \text{conjugate}(F_6)(\phi)$. Equivalently $F_6(X \text{conjugate}(\phi)) = \text{conjugate}(F_6(\phi))$. Thus real coefficients of F_6 plus its conjugate define an invariant interaction on this specified field content. Ordinary componentwise conjugation in the original basis would not establish this result.

There is an important scope restriction. We enumerate all 17 conjugacy classes and the four outer-class representatives supplied by the known Klein outer automorphism group. Every representative fails to send at least one class to its inverse class. For the representative interchanging this triplet with its conjugate, two classes fail. Inner adjustments do not change these class permutations. There is therefore no class-inverting automorphism that could define the same physical CP operation for arbitrary irreducible field content. The anti-linear operation constructed here is restricted to the elementary triplet, singlet and bifundamental fields that pass the explicit consistency and Lagrangian checks. Adding a different elementary

representation requires a fresh audit.

Primary sources: Hagedorn, Meroni and Vitale, <https://arxiv.org/abs/1307.5308> , equation (34), for the outer maps and their classification; Chen et al., <https://arxiv.org/abs/1402.0507> , for the class-inversion requirement in generic finite-group CP; <https://doi.org/10.1140/epjc/s10052-025-15050-8> , for the representation-content dependence of conjugation in Sigma(1080). Our exact class enumeration and intertwiner calculation are independent computations using the recovered generators.

Cross-sector sequestering and its exact operator boundary

$$G_u \times G_d, \quad |\phi_{ui}^\dagger \wedge \phi_{dj}|^2 \quad (\text{degree } 6)$$

Take ϕ_{ui} in (3,1), ϕ_{dj} in (1,3), and Λ in (3,3bar). Retain one independent C6 shaping factor for each right-handed singlet and its flavon. Before a link is inserted, $|\phi_{ui}^\dagger \phi_{dj}|^2$ squared is not an invariant of the product family group. There is no invariant identification of its independent family indices. Its proposed replacement $|\phi_{ui}^\dagger \Lambda \phi_{dj}|^2$ squared is invariant and has field degree six. Its nine coefficients remain independent. Arbitrary additional phase charges could not have forbidden the original absolute overlap square in the diagonal-group model: its charges cancel identically.

The bifundamental Molien calculation averages over the 1080 squared product group using exact power-trace class signatures and Newton identities. Through total degree six it has nonzero spaces (1,1) of dimension one, (3,0) and (0,3) of dimension one, (2,2) of dimension two, (4,1) and (1,4) of dimension one, (3,3) of dimension three, and (6,0) and (0,6) of dimension two. This is the complete self-operator count through that degree, not only a selected contraction.

The degree-three invariant is $\det \Lambda$. The two holomorphic degree-six invariants are its square and $I_6(\Lambda)$, obtained by contracting the unique symmetric sextic tensor with six link matrices and its conjugate tensor. Consequently there is only one genuinely new holomorphic orientation tensor at this degree. Balanced link-only sextics are the three symmetric polynomials of the singular values of Λ . At degree five the invariant is $\det \Lambda$ times $\text{Tr}(\Lambda \dagger \Lambda)$. Real parts are selected by the field-content CP involution; none of these counts supplies a relation between independent coupling constants.

The complete renormalizable scalar count for the listed six flavons, one link and one Higgs is 58 real parameters with this CP restriction. There are eight quadratic norms, one real cubic determinant coefficient, and 49 quartics. The quartics consist of 21 flavon norm products, six within-sector absolute overlaps, two link self-contractions, six link-norm/flavon-norm products, six $\phi \dagger \Lambda \phi$ or conjugate-side contractions, and eight Higgs quartics and norm portals. The nine direct up/down overlap quartics of the earlier diagonal-group model are absent. Without CP the determinant coefficient has two real components, giving 59 parameters. The counts refer to this explicitly assumed product family and shaping content, not to every possible UV theory.

On a unitary link, $\Lambda \dagger \Lambda$ is proportional to I. The six link/flavon orientation quartics then become norm terms and cannot rotate a source frame. Thus this construction removes the previous quartic angular perturbation in that regime. At degree six the dressed overlaps return, and a general Wilsonian effective theory permits them even if a chosen tree potential sets their coefficients to zero. They must be controlled by an additional mechanism before an exact CKM prediction could be protected.

A representation-derived positive link interaction

$$I_6(U) = \langle T, T \circ U \rangle, \quad \text{Re } I_6(U) \leq \|T\|^2 = 16$$

Let T be the symmetric coefficient tensor defined by $F_6(\phi)=T$ contracted with six copies of ϕ . Tensor norms require multinomial weights; the exact polynomial coefficients give $\text{norm}(T)^2=16$. Define $I_6(\Lambda)=\text{inner product of } T \text{ with } T \text{ composed with } \Lambda$. For unitary U , $\text{norm}(T \text{ composed with } U)=\text{norm}(T)$, so $\text{Re } I_6(U)$ is at most 16 by Cauchy-Schwarz. Equality is equivalent to T composed with $U=T$. All 1080 exact group matrices satisfy that tensor identity, independently of the chosen quark model or CKM target.

A globally nonnegative polynomial alignment energy is $V_L = a \text{Tr}[(\Lambda^\dagger \Lambda - \text{ell squared})^2] + b |\det \Lambda - \text{ell cubed}|^2 + c \text{norm}[T \text{ composed with } \Lambda - \text{ell to the sixth } T]^2$, for positive a, b, c and ell . Every $\Lambda = \text{ell } R(g)$ is an exact zero-energy state. On the unitary, determinant-fixed manifold its last term is proportional to $16 - \text{Re } I_6(U)$. The last norm square contains degree-six and degree-twelve terms; the interaction is not a wholly renormalizable potential. Its coefficients and its higher-degree completion are explicit choices. We establish these zero-energy states and their positivity, without claiming an independently derived UV origin or a complete classification of all zero-energy matrices.

The link transforms under the elementary CP operation as Λ maps to $X \text{ conjugate}(\Lambda) X^\dagger$. I_6 and $\det \Lambda$ then map to their conjugates. The positive interaction is therefore invariant. It correlates the sectors through a fixed representation tensor rather than through an arbitrary fitted matrix. Its degenerate group-related states do not select a physical CKM phase by themselves.

A renormalizable messenger origin for the sextic

The sextic also has an explicit interaction-level origin. Symmetric squared of the faithful triplet is an irreducible complex six, and symmetric cubed is an irreducible real ten. The latter has a unique invariant symmetric bilinear Q_{10} . Polarizing the sextic across two groups of three indices gives this bilinear, with $F_6(\phi)=\text{Sym}_3(\phi) \text{ transpose } Q_{10} \text{ Sym}_3(\phi)$. In an orthonormal symmetric-cube basis its exact identity is $Q_{10}^\dagger Q_{10} = (8/5) I$. We certify all 100 entries of that matrix identity in the coefficient field. This establishes nondegeneracy and fixes the internal component coefficients independently of a desired root.

For the product-family link introduce supersymmetric chiral messengers A in $(\bar{6}, 6)$, $\tilde{A}$ in $(6, \bar{6})$, and B in $(10, 10)$, with dimensions 36, 36 and 100. These are Standard Model singlets and neutral under the six shaping factors. The superpotential chain contains $M_6 A \cdot \tilde{A}$, $M_{10} B \cdot B$ divided by two, $y_2 A \cdot (\Lambda \Lambda)$ projected to the 36, and $y_3 B \cdot (\tilde{A} \Lambda)$ projected to the 100. Every displayed family contraction is unique. The CP action on these messenger fields is inherited from the symmetric-square and symmetric-cube actions of the already checked X ; no arbitrary new CP matrix is introduced.

Eliminating the massive fields at leading tree order generates a holomorphic operator proportional to $-(y_2 y_3)^2 I_6(\Lambda)/(2 M_6^2 M_{10})$, in compatible invariant normalizations. Thus the sextic is generated by renormalizable cubic vertices whose representation coefficients fix its component pattern. This is stronger than simply inserting a high-degree polynomial into the scalar potential. It is a supersymmetric messenger-sector construction, not a completed softly broken scalar model: the F-term scalar potential from the effective superpotential, soft terms, canonical kinetic corrections and physical vacuum selection must be matched separately.

The full audit counts all 34 possible monomial multisets of degrees one through three for these four chiral field species. There are no invariant linear terms, two independent mass contractions, and 15 cubic contractions: the determinant of Lambda; the two unique chain vertices; and four self-cubic contractions each for A, A_tilde and B. The unused self-cubics are not silently excluded. A_tilde self-cubics can contribute at degree six to the same two-dimensional link-invariant space. The A and B self-cubics first affect higher orders in the small-link expansion for this massive branch. Hence the generated orientation tensor is fixed, while its coefficient and the determinant-squared coefficient remain independently adjustable. Other branches and the regime of large link expectation values require the full messenger equations.

The extra messenger fields do not couple holomorphically to the charged source flavons at renormalizable degree under the listed shaping charges. This controls a concrete set of interactions, while higher-degree Kahler and effective scalar terms can still generate the dressed cross-sector overlaps. No radiative absence theorem for those overlaps is asserted.

Complete mediator matching with the reverse interaction included

Assign Q_L to (3,1). The up-sector mediator remains a triplet of G_u . For the down sector introduce D_u in (3,1) and D_d in (1,3), both vectorlike Standard Model weak singlets of down electric charge. The right-handed down singlets couple to D_d through the three down flavons. The allowed link Yukawa couplings run in both directions: $g \bar{D}_{uL} \Lambda D_{dR}$ and $g_{\text{reverse}} \bar{D}_{dL} \Lambda^\dagger D_{uR}$. Both are allowed by the same mass and shaping charges. Omitting the reverse contraction would incorrectly restrict the operator space.

At $H=0$ the down heavy block is $[B, M]$, with $B=[0; C_d]$ and $M=[M_u I, g \Lambda; g_{\text{reverse}} \Lambda^\dagger, M_d I]$. The light normalized right frame is $[I; -A] K$ to the power $-1/2$, where $A=M^{-1}B$ and $K=I+A^\dagger A$. The Higgs couples Q_L to D_{uR} , giving $Y_d=-h A_{\text{top}} K$ to the power $-1/2$. For an invertible M there are exactly three right light modes and six massive down mediator modes.

For $\Lambda=\ell U$ and real parameters, write $D=M_u M_d - g g_{\text{reverse}} \ell^2$. Then $A_{\text{top}}=-g \Lambda C_d/D$ and $A_{\text{bottom}}=M_u C_d/D$. Consequently $K=I+[(g^2 \ell^2 + M_u^2)/D^2] C_d^\dagger C_d$, and Y_d is a scalar times $U C_d$ times that canonical factor. The left frame is exactly U times the left singular frame of C_d . This includes the reverse link coupling and all orders in flavon mixing. The heavy masses must remain nonzero, requiring D nonzero. Finite-Higgs and quantum threshold corrections are separate calculations.

Physical CP established through quark invariants

$$J \neq 0, \quad \text{Im det}[H_u, H_d] \neq 0, \quad J_{\text{CP}} = -J$$

The numerical example uses a unitary W spanning the real fixed space of the anti-linear map, so $X \text{conjugate}(W)=W$. Its source matrices are $C_u=W \text{diag}(0.07, 0.31, 0.9)$ and $C_d=W \text{diag}(0.08, 0.27, 0.85)$. All listed mediator couplings are real. We evaluate every group link $U=R(g)$, select the largest absolute J without a target phase or mixing anchor, and calculate the conjugate link $X \text{conjugate}(U) X^\dagger$.

After canonical matching the pair has $|J|$ approximately 0.0703201663, identical mixing magnitudes and opposite J signs. The chart phases are approximately plus and minus

94.6111562 degrees, and C_{eff} is approximately 1.50313413. An independent weak-basis check gives nonzero $\text{Im det}[H_u, H_d]$, approximately $7.09\text{e-}9$ in absolute value. Nonzero values with nondegenerate quark spectra demonstrate physical CP violation; complex representation entries or flavon phases alone would not do so.

This example is conditional on the chosen CP-real source frames and on a selected group link vacuum. The CP-real basis itself has a real orthogonal freedom, and the scalar model has not selected those frames. The large mixing is not phenomenologically viable. Real, independently allowed column-coupling changes in a nonorthogonal source example alter both C_{eff} and the physical phase. The receipt retains all cases. Thus physical CP is possible within a consistent restricted elementary symmetry, but the observed phase and the golden relation remain unpredicted.

The canonical global spectrum is gapped

The recovered six-dimensional real cap fiber is wedge squared of the standard four-dimensional A5 representation, with its stored positive metric. Construct the covariant graph Laplacian using the actual 252 directed incidence transports on a 42-cap component. In integral coordinates its characteristic polynomial is $(t-2)^{15} (t-3)^{55} (t-5)^{21} (t-6)^{70} (t-7)^{21} (t-9)^{55} (t-10)^{15}$. We compute and factor this polynomial exactly over the integers. The numerical metric-whitened Hermitian matrix independently reproduces the spectrum.

The base connection has dimension 252 and zero-dimensional kernel. Its orientation double cover has dimension 504 and the same spectral levels with doubled multiplicities: the other block is obtained by t mapping to $12-t$. Both have exact mass-squared gap two. On all 176 isomorphic cap components the base space contains 44,352 modes and has no zero mode. A parallel rank-three projector therefore does not yield three parallel vectors. The A5 cover holonomy has no invariant vector in either irreducible golden triplet, consistent with the spectral result.

A global three-mode operator with explicit breaking

$$\dim \ker M^2 = 3, \quad \lambda_{\text{next}}(M^2) = \frac{1}{2}, \quad N = 44\,352$$

To exhibit exactly what an additional assumption can accomplish, choose one 42-cap component and a spanning tree, transport a root frame along that tree, and replace every edge matrix by the corresponding pure-gauge transport $R_w R_v$ inverse. This keeps the tree transports and changes 160 of the 252 directed edge matrices of the component. It trivializes the holonomy and is an explicit change to the canonical connection. It is not invariant under the full recovered M22 incidence action.

Transport the root rank-three Hodge projector by the same frames, and add a one-half penalty on its orthogonal complement. The scalar component Laplacian has exact spectrum 0 with multiplicity one, 4 with multiplicity 21, 7 with multiplicity six, and 9 with multiplicity 14. The selected six-dimensional bundle therefore has three zero modes, and levels one half, 4, 4.5, 7, 7.5, 9 and 9.5 with multiplicities 3, 63, 63, 18, 18, 42 and 42 respectively. Gap every other cap fiber by an onsite mass squared of two, adding 44,100 massive modes.

This explicitly defined global mass-squared operator has exactly three zero modes among 44,352 modes and gap one half. Their normalized wavefunctions are transported root vectors divided by $\sqrt{42}$; the numerical mass and normalization residuals are below $2\text{e-}15$. The spectral statement is exact from the scalar characteristic polynomial and pure-gauge similarity. Selecting

the component breaks M_{22} , and flattening its connection breaks further symmetry. We have not derived either selection dynamically. The operator is a positive Hermitian flavor mass-squared construction, not a gauge-invariant chiral quark UV completion. Its real A_5 transport and the local complex Valentiner link model have not been unified into one global particle theory.

What remains after this push

The bare quartic deformation is now excluded by specified product symmetry, but the degree-six dressed operators survive. The CP action is explicit on the listed fields, and physical CP is demonstrated in canonically matched matrices, but its vacuum and numerical phase are not selected. The global spectrum is calculated, including an exact obstruction for the canonical connection and a conditional three-mode alternative, but the required breaking interactions and chirality have not been derived. These boundaries are sharper than another constructed golden root. A complete predictive theory must connect the product-family link, its mixed operator coefficients and CP-compatible scalar vacua to a dynamically selected global chiral spectrum.

Reproduction and receipts

Run `python python/develop_valentiner_cp.py`, `python python/develop_valentiner_link.py` and `python python/develop_m22_global_modes.py`, and `python python/develop_valentiner_messenger.py`. The computations use the existing exact triplet generators, sextic receipt and frozen cap transports. The new receipts are `valentiner_cp.json`, `valentiner_link.json`, `m22_global_modes.json` and `valentiner_messenger.json` under `receipts/m22_interactions`. The CP and polynomial identities, class map propagation, bifundamental invariant counts and characteristic-polynomial factors use exact arithmetic. The quark examples, matrix whitening and normalization residuals are numerical cross-checks. No experimental data or new Lean theorem is claimed.